



**A FIELD REPORT ON
TRAVERSING, HYDROPOWER SURVEY, BRIDGE
SURVEY, ROUTE SURVEY AND PLANE TABLE
SURVEY**

A THIRD YEAR PROJECT SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENT FOR
THE DEGREE OF BACHELOR OF ENGINEERING
IN
GEOMATICS ENGINEERING

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ABSTRACT

This survey field report outlines the use of various engineering and construction surveying techniques in the area of Jiri, Dolakha district. The surveying methods used in the project were diverse and suited to the requirements of the terrain and the purpose of the project. Those methods were traverse, hydropower component surveys (Intake, Powerhouse, Canal, Penstock, and Tailrace), linked traverse survey, route survey, bridge survey and plane table survey.

This report provides the objectives, scope, methodology of the project, observations, and calculations used to establish vertical and horizontal control points and create a topographic map using different surveying techniques during the field work. Traversing was used to determine the angles and distances between points to determine their location as control points in future engineering surveys.

All in all of the surveying methods allowed us to determine the locations of the important engineering elements in the Jiri Khola region correctly. The obtained data will be used in the future projects of the region and will be used to develop the sphere of land surveying.

Location: Jiri Khola, Dolakha District, Bagmati Province Nepal.

Duration: 3th November, 2025 to 30th November 2025 (28 days)

Working Time: 8:00 am to 4:30 pm

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LIST OF ABBREVIATIONS

ALP	Alignment Line Point
AutoCAD	Automatic Computer-Aided Design
Ccg	Centi Centi Grade
Cg	Centi Grade
CSV	Comma Separated Values
E	Easting
EIA	Environment Impact Assessment
HC	House Corner
HFL	High Flood Level
HS	Horizontal Scale
IP	Intersection Point
L-Section	Longitudinal Section
m	Meter
MW	Megawatt
N	Northing
PNEZD	Point Northing Easting Zenith Description
PTS	Plane Table Survey
RC	Road Center
RCC	Reinforced Cement Concrete
RL	Reduced Level
RL (road)	Road Left Edge
RO	Reference Object

ROR	Run of River
RR	Road Right Edge
SH	Spot Height
SWDTM	Softwel Digital Terrain Model
TS	Total Station
U/S & D/S	Upstream and Downstream
VS	Vertical Scale
WGS	World Geodetic System
X-section	Cross-Section

CHAPTER 1. TRAVERSING

1. INTRODUCTION

1.1 Background

Traversing is the type of survey in which the length and direction of a series of connected lines are measured with the help of distance and angle measuring instruments respectively. Generally, the linear measurements are made with the help of tape, chain or EDM, and the relative direction of the lines are measured with a compass or theodolite. It is a technique of establishing horizontal control points in dense and narrow areas where second order triangulation is not possible.

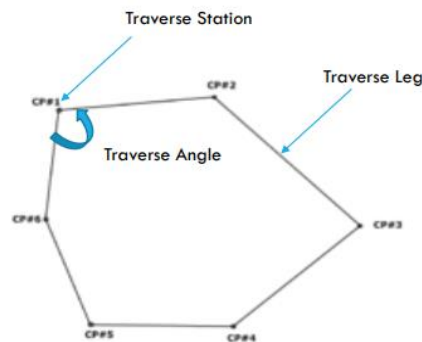


Figure 1: Traverse

Traversing has a wide application in various fields of geospatial survey procedure. Some of the main purposes of traversing incorporate determining existing boundary lines, calculation of area within the boundary, establishing control points for photogrammetric surveying and mapping, establishing control points for calculating earthwork quantities and so on.

Principle of traverse surveying:

The principle of traversing is that if the direction and length of any line is known then the coordinates of its head (end point) can be found from the given coordinates of its tail (starting point).

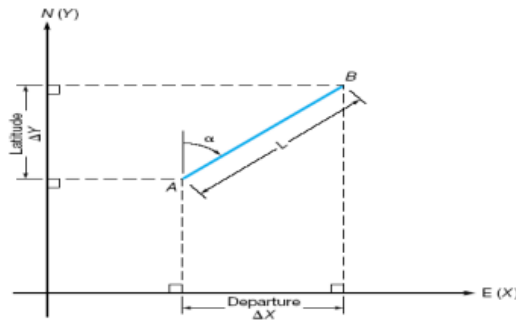


Figure 2: Latitude and Departure in traversing

$$\text{Latitude } (\Delta N) = d \cos \theta$$

$$\text{Departure } (\Delta E) = d \sin \theta$$

If the coordinate of A is known.

Then the coordinate of B can be calculated.

$$\text{i.e. } E_2 = E_1 + \Delta E \text{ and } N_2 = N_1 + \Delta N$$

1.2 Types of traverse surveying

Traverse surveying is classified on the basis of following headings:

- A. On the basis of starting and ending point
- B. On the basis of instrument used
- C. On the basis of accuracy and precision

A. Traverse based on starting and ending point:

Further on the basis of starting point and ending point, traverse can be divided into 2 divisions:

a. Closed loop traverse:

It starts either from a known starting point and ends at the same station. It has more checks and it is thus more accurate. It is itself a linear and angular check. It is suitable for the survey of boundaries of ponds, sports grounds, forests, etc.

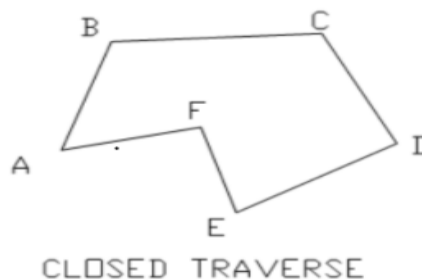


Figure 3: Closed traverse

b. link traverse:

It either starts from a known or unknown station and ends at another unknown station or known station. It is geometrically open but mathematically closed. It checks only possible by additional observation. It has no check itself and is used in linear alignment. It is mostly suitable for long narrow strips. For example: it is used for surveying of roads, coastal lines.

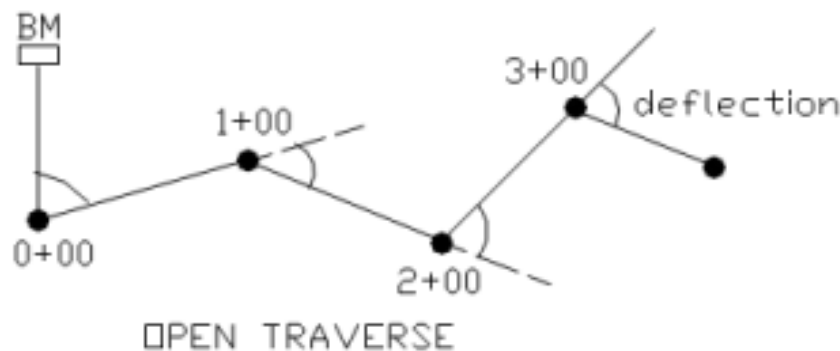


Figure 4: Open traverse

B. Traverse based on instrument used:

a. Chain traverse

In this traverse, only linear measurements are taken. Angles are preserved or set out by a tie line. It is mainly used for flat, level and small areas.

b. Compass traverse

Magnetic Compass is used for measuring direction of leg and tape is used for distance measurement. It is used for small, busy ground where chain traverse is not possible. Bowditch's method is used to adjust traverse.

c. Plane table traverse

In this traverse, alidade is used for direction of traverse leg and distance measured by tape. It is used for a comparatively larger area than a compass. In this traverse, adjustment is done graphically.

d. Theodolite traverse

In this traverse, theodolite is used for angular measurement and steel band is used for linear measurement. It helps in precise and accurate control point establishment. The adjustment in this traverse is done by Transit rule/Bowditch rule.

e. Tacheometry traverse

In this traverse, angle measurement is the same as in theodolite traverse and the distance is measured by stadia method.

C. Traverse based on accuracy and precision:

a. Primary traverse:

Primary traverse is highly precise traversing. Long range EDM is used for distance and precise theodolite T2/T3 is used for angle measurement in this traverse. It is used where second order triangulation is not possible. In this traverse, 15km of traverse legs are taken into account.

b. Secondary traverse:

Secondary traverse is conducted under controlled area of primary traverse. It is used to establish the third and fourth order points. In this traverse, 5km of traverse leg is taken into account.

c. Tertiary traverse:

Tertiary traverse is conducted under controlled area of primary and secondary order traverse. It is used for cadastral purposes.

1.3 Objectives

The main objective of traversing project was to establish horizontal and vertical control points of the fourth order tacheometric surveying around Jiri area using theodolite.

The secondary objectives achieved during the accomplishment of traversing are pinpointed below:

- To provide the framework for taking details to prepare the Topographic maps.
- Learned about measuring distance, bearing and interior angles of the traverse leg.
- Learned about setting up the monuments in the field.
- Preparation of D cards
- Measurement of horizontal and vertical angle using theodolite.

1.4 Scope of work

The course of traversing was basically focused on establishing control points. It dealt with the scope of determining cultural features of the traverse stations (i.e. (x, y, z) coordinates). We had established 13 control points by the method of traverse which can be further used for densification of the control points. The monumentation used for

stations are of temporary nature (wooden pegs are used). So, permanent monumentation of the control point is not under the scope of the project. We carried out theodolite traversing. Hence, traversing by Plane Table, Chain and Compass is not under the scope of our project. Furthermore, preparation of D-Cards of monuments is not under the scope of the project.

2. METHODOLOGY

2.1 Study area

The suitable site that was selected for traverse was around Jiri Airport area which is shown in map below:

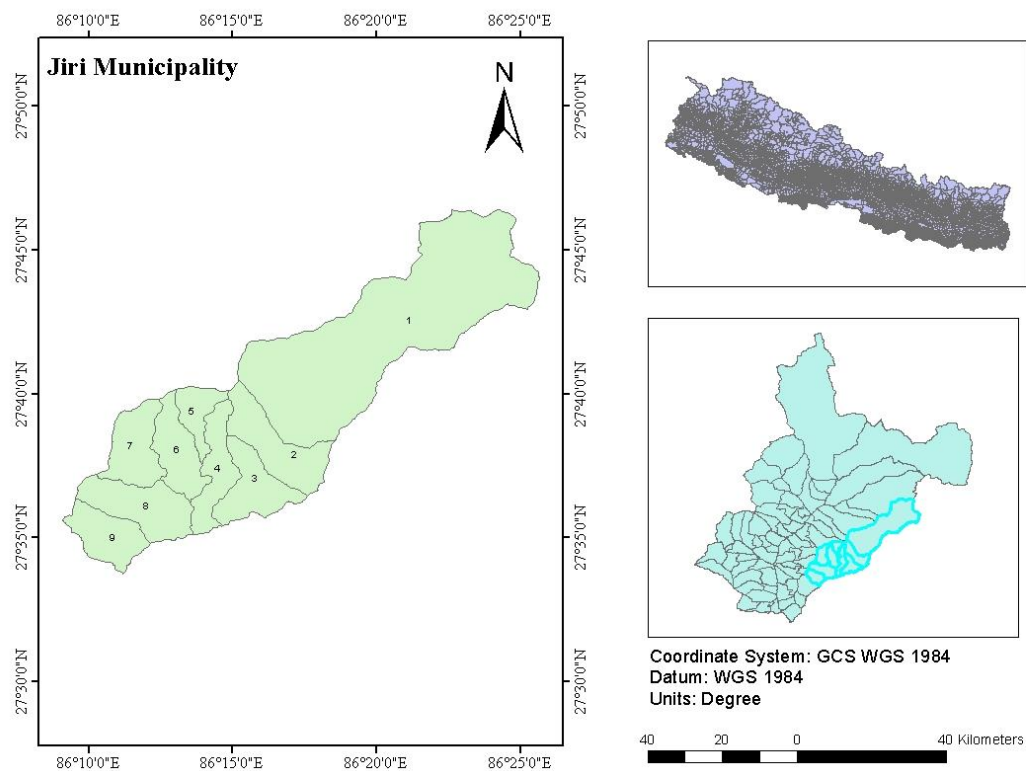


Figure 5: Study Area Map of Jiri, Dolakha

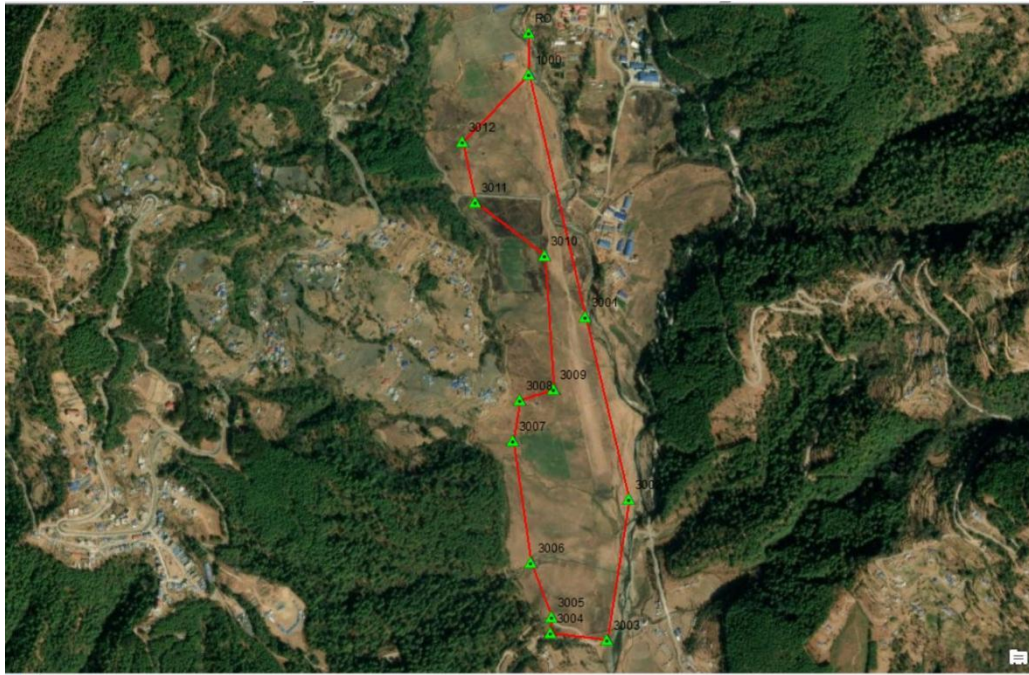


Figure 6: Study area of Closed Traverse surveying

2.2 Instrument used

- Total Station

A total station is a surveying instrument that measures distances, angles and heights to create maps, construction plans and 3D models. A total station combines an electronic theodolite and an electronic distance meter (EDM). The electronic theodolite gives angles (horizontal and vertical).

- Tripod

A tripod is a three-legged stand used to mount a total station on it. It has movable legs to make setting of TS easier.

- Prism (Reflector)

The prism also called reflector reflects back the electromagnetic wave emitted by the EDM.

- Prism Pole

Prism pole is used to mount prisms and also for signalling. It has a level bubble in it to ensure the rod is levelled in a vertical position.

- Measuring tape

A 30m fibreglass tape was used to measure distances or heights such as HI of the instrument.

2.3 Software Used

The software used in traverse are:

- i. MS Excel
- ii. ArcMap 10.8.2

2.4 Specifications

Particulars	Specification
Types of traverses	Closed traverse
Order of the work	Fourth order
Numbering	1000-1013
Time required for traverse	3 days
Name of instrument	NIKON
Purpose	To establish Horizontal and Vertical control points
Unit of Measurement	Linear measurement: Meter Angular Measurement: Grade
Instrument used	Linear Measurement: Total Station Angular measurement: Theodolite
Linear Measurement Direction	Two-way, double measurement
Angular Measurement Direction	Clockwise
Number of sets for angular measurement	Three sets
Face to Face tolerance	2 cg
Set to Set tolerance	60 ccg
RO to RO	30 ccg
Angular Accuracy	$75\sqrt{n}$ (n is number of traverse station)
Method of Angular Misclosure adjustment	Equal shift
Method of Linear Misclosure adjustment	Bowditch rule
Method of vertical Misclosure adjustment	Equal shift

Table 1: Specification for traverse surveying

2.5 Methodological Workflow

- **Reconnaissance:** Reconnaissance is fundamentally the first field operation and prominent stage. In this phase, we fully inspected the area to be surveyed. We marked the position of stations on the ground with wooden pegs. There was a total of 12 stations for three loops.
- **Monumentation:** In this phase, we nailed the station with wooden pegs on ground. As there were three loops, the station number was written on each peg.
- **Signaling:** Some of the monuments that we marked on the stations couldn't be seen from nearby stations. We employed range rods set up vertically above the stations centers as a signal during traverse.
- **Observation:** As there were 12 stations, we recorded all horizontal angles, zenithal angles, and traverse leg distance.
 - a. **Horizontal angles:** For the observation of horizontal angles, we took a total of 3 sets of readings (angles set as: $000^{\circ}10'00''$, $067^{\circ}10'00''$ and $133^{\circ}10'00''$) to eliminate the graduation error. For each set we also took reading from each face (both right and left face) to eliminate the collimation error from the Total Station. To change the face of TS, the method swinging (rotating the TS by 360° in horizontal plane through vertical axis) and transiting (rotating the TS by 180°). All the errors were within the tolerance level.
 - b. **Traverse leg distance:** In order to take the length of each traverse leg, we used the Total Station as it gives the distance with higher precision. We took the average of readings from both sides to get precise distance.
 - c. **Zenithal angle:** For the measurement of zenithal angles, we used TS. Zenithal angle is widely preferred over the vertical angle. As zenithal and vertical angles are compliment of each other we can easily calculate both easily for any one angle. In the field, we took 1 set of zenithal angles for each station (both faces). These zenithal angles were used for calculating RL of each station.
- **Computation:** The tolerance calculations were done on field whereas the traverse form calculations were done on returning to room. Some formulas that were used for traverse calculation are mentioned below:

Angular misclosure = Calculated value of Sum - Sum of interior angles of a

Polygon. Closing error = $\sqrt{[(\text{sum of latitude})^2 + (\text{sum of departure})^2]}$

Relative closing error = closing error/perimeter of traverse

Direction of closing error (A) = $\tan^{-1}$ (sum of departure/sum of latitude).

Also, the errors were adjusted, and the angular adjustment technique used is described below:

a. Angle Adjustment Method:

As the total number of stations was 12 and we observed angles at each station. It should be noted that errors are inevitable in nature so certain corrections must be made before final traverse calculations.

$$\begin{aligned} \text{Total sum of horizontal angles} &= (2n + 4) * 100^g \\ &= (2*13+4) * 100^g = 3000^g \end{aligned}$$

Where, n is total number of stations.

But errors accumulated due to systematic, natural, and random errors, the total sum of angles was found to be $3000^g00^c72^{cc}$ with angular misclosure of $000^g00^c93^{cc}$ which is within the tolerance for angular misclosure given be:

$$\text{Permissible Misclosure} = k\sqrt{n} = 75\sqrt{13} = 270^{cc}$$

Where, k is least count and n is total number of angles.

Now, the error was adjusted by *Equal Shift Method* in each station. If the sum of the measured angles does not add up to the geometric value $\{(2n+4) * \text{right angle}\}$ the difference is divided by the number of exterior angles, and the result is then distributed. This correction is either added to or subtracted from each measured angle so the sum will become the true value. The total error of $000^g00^c72^{cc}$ were distributed in 13 readings with each reading getting the correction of about $+000^g00^c06^{cc}$. The correction applied was positive because the error accumulated was negative.

b. .Bearing Computation Method:

The coordinates and RL RO and station 1000 were already provided. Station 3001 was common for all groups. From these coordinates the bearing of line 1000-3001 was calculated and then using proper formula the bearing of other lines was also derived.

$$F.B = B.B \text{ of Previous line} + \text{Included Angle}$$

$$B.B = F.B \pm 200$$

Where, B.B is back bearing and F.B is fore bearing of lines.

- Result of Observation

Close loop traverse and coordinates were given as RO and 1000 respectively. link traverse was provided between stations RO, 1000, 1001 and RO2.

S.N.	Easting	Northing	Height	Remarks
1	423958.865	3056787.54	1852.005	1000
2	423958.865	3056888.5	1857.265	RO

Table 2: Known coordinate for closed traverse

S.N.	Easting (m)	Northing (m)	Elevation (m)	Remarks
1	423926.849	3056804.454	1900.665	RO1
2	423998.434	3056690.833	1890.649	1000
3	423822.626	3055052.227	1964.502	1001
4	423851.707	3055033.912	1962.837	RO2

Table 3: Known coordinate for link traverse

3. RESULTS

The above-mentioned processes were done in order to find the coordinates of each traverse station by using formula and trigonometry. The angles and distances were used to find the ΔX (difference in X) and ΔY (difference in Y) from the known point. By using simple operations like addition and subtraction, the coordinates of new points were calculated from given coordinates of station RO and 1000. And the coordinates of each station that we calculated are given below:

Line	Easting(m)	Northing(m)	Elevation(m)
RO-1000	423958.865	3056888.5	1857.265
1000-3001	423958.865	3056787.54	1852.005
3001-3002	424094.4583	3056200.966	1814.627
3002-3003	424200.7671	3055758.45	1800.056
3003-3004	424147.4655	3055417.999	1795.293
3004-3005	424010.2647	3055436.25	1824.084
3005-3006	424012.4938	3055474.131	1813.251
3006-3007	423963.8114	3055605.122	1810.441
3007-3008	423919.8042	3055901.086	1818.458
3008-3009	423936.8354	3055998.483	1832.828
3009-3010	424017.5248	3056024.021	1815.693
3010-3011	423996.6197	3056348.075	1822.63
3011-3012	423829.1956	3056477.248	1839.226
3012-1000	423797.6312	3056625.285	1852.626
1000-RO	423958.865	3056787.54	1852.005

Table 4: Coordinate obtained from closed traverse

4. CONCLUSION

As the starting of our field, we completed traverse within 2 days where we observed the horizontal angles, zenithal angles, bearings of line and traverse leg distances for the calculation of coordinates and RL of each station. The total perimeter(p) for our traverse loop was 3116.873m and error(e) was 0.149m. And the final linear accuracy obtained was calculated as follows:

$$\text{Linear Accuracy} = \frac{1}{\frac{p}{e}} = \frac{1}{\frac{3116.873}{0.149}} = 20866.657$$

Where,

P is the total perimeter of traverse leg whereas e is the error.

We got to know about the setting up of control points following adequate reconnaissance, the significance of intervisibility of stations and teamwork. We also learned how to use all the instruments in this project properly. We would suggest that surveyors should treat the instruments correctly and ensure that they are oriented correctly and adjustment of TS.

CHAPTER 2. HYDROPOWER SURVEY

1. INTRODUCTION

1.1 Background

The transport of water has been an important source of mechanical and later electrical power since ancient times. Through the kinetic and potential energy of moving or falling water, hydropower systems can transform the natural movement of water into a source of clean energy that is sustainable and continuous (Boyle, 2012). Water power, also known as hydropower, is a type of power that utilizes the power of flowing or falling water to power a machine or to produce electricity by converting the gravitational potential or kinetic energy into useful power (Paish, 2002).

Hydropower has become the primary source of hydroelectric power generation and is an important source of pumped-storage hydroelectricity, a major technique of large-scale energy storage and grid stabilization (Twidell and Weir, 2015). The advantages of hydropower have been known since time immemorial; in addition to being a clean and inexpensive source of renewable energy, hydropower plants can quickly provide electricity to the grid, which is why they are a very reliable option of backup power in case of an outage or disruption (Kaunda, Kimambo, & Nielsen, 2012). Also, hydropower systems help in flood control, irrigation and management of water supply, which adds to the wider environmental and socio-economic advantages.

Hydroelectricity is one of the most important sources of renewable energy in the world, and in recent years, it provides about 15 percent of the electricity in the world (International Renewable Energy Agency, 2021). It is done by utilizing the energy of water to rotate turbines to produce mechanical energy that is converted to electricity and thus hydropower is a highly used and sustainable power technology.

Hydropower infrastructure development needs to be based on comprehensive feasibility studies to determine geological, hydrological, and topographical factors. The surveying of hydropower is necessary in determining the appropriate sites where dams, reservoirs, powerhouses, penstocks, and other parts of the station will be located (Chow, Maidment and Mays, 2010). In this regard, sixth-semester Geomatics Engineering students had to conduct a hydropower survey to model a hydropower system and identify possible locations of all the required elements of a hydropower station.

1.2 Objectives

1.2.1 Primary objective

The main aim of the hydropower survey is to make a comprehensive topographic map of the potential site so that it is possible to identify, assess, and select the appropriate location of the different parts of a hydropower plant, including the dam, intake, spillway, headrace, penstock, powerhouse, and tailrace.

1.2.2 Secondary objective

- To determine the alignment of the canal and at the same time provide the right gradient and hydraulic efficiency.
- To establish and correct the best alignment of the penstock based on slope, terrain, and structure.
- To produce comprehensive longitudinal profiles and cross-sections of the canal and the penstock alignment.
- To carry out discharge measurements to determine the available flow and aid in the hydropower feasibility analysis.

1.3 Theoretical Framework

The hydropower survey is based on the principles and procedures needed in planning and building of a hydropower station. Such infrastructure development is dependent on the accurate surveying and mapping to guarantee the correct site selection and the maximum positioning of all structural elements. It starts with a control survey of the area in question and creates a network of control points that are well distributed and which are used as the basis of the further detailed surveys.

After drawing the control points, a thorough surveying is conducted between the intake area and the powerhouse location. A detailed topographic map is then drawn using these fixed control points as the reference points to aid the design and feasibility study of the hydropower project.

One of the major components of this process is traversing, which consists in developing a network of lines of the survey whose directions and lengths are systematically measured. Historically, theodolites and chains or tapes were used to measure angles and distances respectively. In the current hydropower surveying, however, Total Station traversing is used in order to obtain increased speed and accuracy. A Total Station is a

combination of Electronic Distance Measurement (EDM), a theodolite and onboard computing capabilities, enabling angular and linear measurements to be made with one tool. This integration guarantees the uniformity of the accuracy, the increase in the data reliability, and the simplification of the overall surveying process.

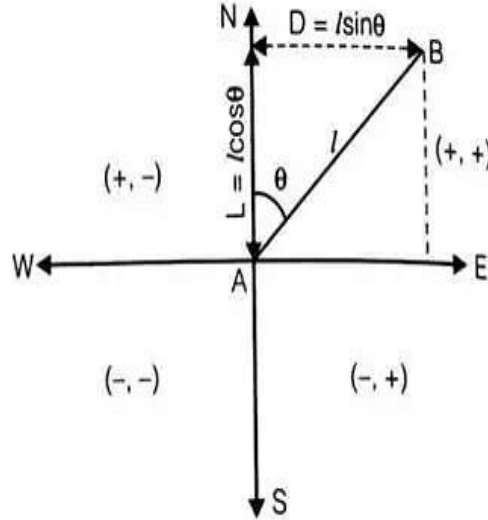


Figure 7: latitude and departure in traversing

Here,

Latitude is given by, Latitude (ΔN) = $L \cos \theta$

And, Departure (ΔE) = $L \sin \theta$

As according to the principle, if the coordinates of point A is known, i.e. A (E_1, N_1)

Then, the unknown coordinates of B (E_2, N_2) can be computed as;

$E_2 = E_1 + \Delta E$ and $N_2 = N_1 + \Delta N$.

1.4 Scope of work

A hydropower survey scope of work includes site assessment and systematic data collection that is required to determine the technical and environmental feasibility of the project. The survey procedure involves the topographic mapping of the area, river courses, and elevation profiles with the help of such tools as GPS, Total Stations, and leveling tools. These measurements are used to determine appropriate sites of important hydropower structures such as the intake, canal, powerhouse, penstock and tailrace.

Besides topographic mapping, hydrological surveys are also performed to examine the features of the streamflow, the discharge rates, and the seasonal changes of the water availability. These hydrological parameters are necessary to estimate the possible power generation capacity and guarantee a long-term operational reliability.

Both topographic and hydrological surveys provide the data that is used in the engineering design and structural analysis, capacity estimation, and general feasibility evaluation. The hydropower survey will help in ensuring that the proposed project is not only sound technically but also sustainable and structurally viable through provision of accurate and reliable information.

1.5 Some Terms Used in Hydropower Survey

- a. **Alignment:** The line joining the intersection points is known as alignment.
- b. **Control Points:** They are the points located on the ground by traversing that are used for further detail survey.
- c. **Downstream Site:** The area in the hydropower plant below the intake site is known as downstream area.
- d. **Grading:** It is the ratio of vertical distance and horizontal distance between any two points. It is an important factor during canal alignment fixation.
- e. **Head:** The elevation difference between the forebay and powerhouse is known as head. The energy generation is mainly depending on the head factor.
- f. **High Flood Level:** A high flood level is the maximum level to which a water could rise due to rainwater and runoff during a flooding event.
- g. **Strip Survey:** Survey done along the alignment is known as strip survey. It is basically carried out in canal, penstock, tailrace etc.
- h. **Upstream Site:** The area in the hydropower plant above the intake site is known as upstream area.
- i. **Water Level:** A water level is the normal water level to which stream normally flows.
- j. **Weir:** Weir is a solid structure put across a river to raise the water level and divert the water level into the canal.

2. COMPONENTS OF HYDROPOWER

Major components of a hydroelectric plant are:

1. Forebay

A temporary storage basin that is situated above the intake structure of a hydropower plant is referred to as a forebay. It acts as a controlling reservoir in which water is stored

and then it flows into the intake chamber. The forebay capacity depends on the demand and operational conditions of the water that is required and the conditions of the plant. It also acts as a buffer in times of low load, when there is surplus water that can be stored in the meantime.

In most hydropower systems, particularly where large dams are built, the forebay may be the main reservoir above the dam. The reservoir is used as the forebay with water stored in the reservoir being carried to the powerhouse by penstocks.

2. Intake

The intake structure is a collection of the water in the forebay or reservoir and directs it to the penstock system. It is made to control the water inflow, avoid debris that may enter the system, and also to assure safe and efficient flow to the turbines. There are a number of intake structures that are available and the design of the structure varies depending on the site conditions, including terrain, hydrology, sediment load, and structural needs.

3. Penstock

The penstocks are huge and pressurized pipes that transport water through the intake structure or reservoir to the turbines. They are laid with a gradient, to keep the water flowing, and since they are run under pressure, a sudden opening or closing of the penstock gates may cause a water hammer effect, or sudden rush of pressure, which may break the pipe unless well checked.

4. Surge chamber

A surge tank is a cylindrical, open top tank, which is attached to the penstock usually close to the powerhouse, which assists in the regulation of pressure variations in the penstock. In case the powerhouse decreases or denies the water flow, the water level in the surge tank increases, and the surplus pressure is absorbed, which prevents possible damage caused by water hammer.

5. Hydraulic turbines

Hydraulic turbines are machines that harness the hydraulic power of moving water to mechanical power which is then converted to electrical power by a generator attached to it. The high-pressure water in the penstock impounds the turbine blades or runner, and

the central shaft is turned. This is a rotating mechanical energy that operates the generator to generate electricity.

6. Power house

The building that contains and secures the hydraulic and electrical equipment of a hydropower plant is called powerhouse. All the large machinery, such as turbines and generators, is placed on a robust foundation or substructure to hold the equipment and provide safe and efficient operation.

7. Tailrace

This is the channel or passage that water moves out of the turbines and back to the river or another forebay. Preferably, the powerhouse should be near the stream to reduce construction of tailraces. In the event that the powerhouse is located further, a special line should be constructed to transport water safely back to the stream.

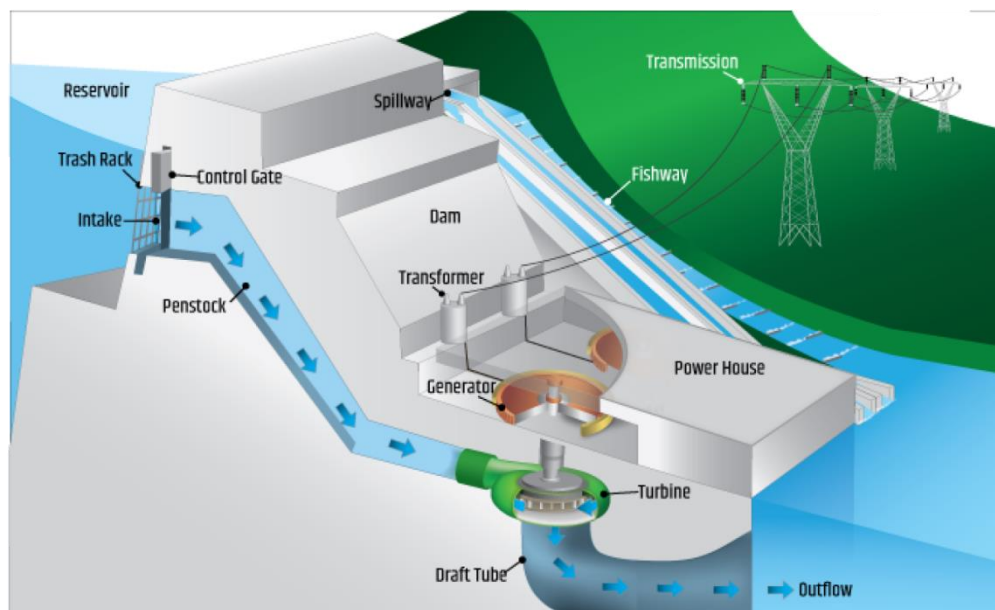


Figure 8: General hydropower components

3. METHODOLOGY

3.1 Study Area

The study area for hydropower survey was selected inside the premises of Jiri River and the surrounding hills around Jiri Airport in Jiri Municipality, Dolakha.

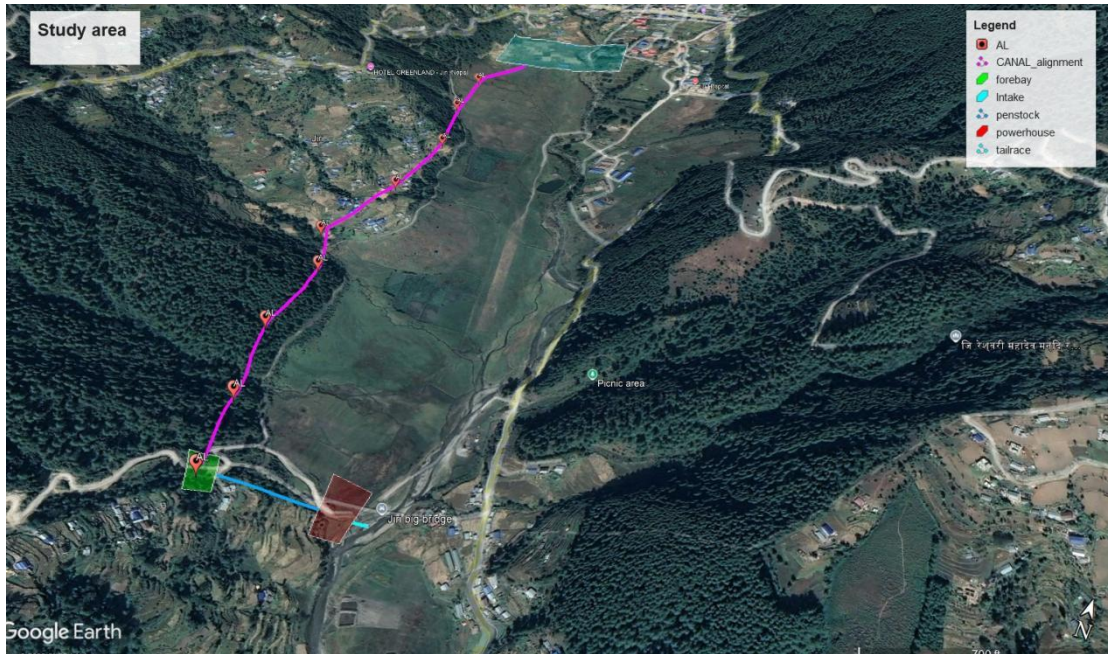


Figure 9: Study area for hydropower survey

3.2 Specification

Particulars	Specifications
Order of the work	Fourth Order
Purpose	To establish control points and perform a detailed topographical survey.
Instrument	NIKON TS
Extent for Reservoir Area (Intake)	300m*300m (for upstream) 200m*100m (for downstream)
Details	stream centerline, water level, HFL, other

	natural and man-made feature
Canal Gradient/ Tunnel Gradient	1 in 500 (approx. 2%) Strip of 50m (25m in each side of alignment) Cross Section, Upstream 25m, Downstream 25m
Tailrace Gradient	1: 100
Penstock	Location of start and end of penstock Strip of 50m (25m in each side of alignment)
Powerhouse	200m*300m area Includes tailrace
Contour Interval	1 m
Scale	1: 1000

Table 5: Specification of preliminary survey

3.3 Instrument and Software Used

Total station: A Total Station is a tool used in surveying and construction to measure distances and angles. It works using a built-in telescope and an electronic distance meter. It also has a small computer that stores the measurements. For our hydropower survey, we used a Nikon Total Station.



Figure 10: Total station

Measuring Tape: A measuring tape is used to measure straight-line distances. In our survey, it was used to measure the height of the instrument and to prepare D-cards by measuring the distances between permanent features and the survey station.



Figure 11: Measuring tape

Prism: A prism, also called a reflector, is used to receive and reflect the electromagnetic waves sent by the Total Station. During the survey, it was attached to prism poles and used for collecting measurements.



Figure 12: Prism

Tripod: A tripod is used to hold the Total Station steady above the survey point.



Figure 13: Tripod

Prism Pole: Prism pole is used to mount prisms and also for signalling. It has a level bubble in it to ensure the rod is levelled in a vertical position.



Figure 14: Prism pole

AutoCAD: AutoCAD is a computer program used to create 2D and 3D drawings. It is a design software developed by Autodesk.

ArcGIS: ArcGIS was used to create a map of the survey area.

Microsoft Excel: Excel was used to bring in survey points from the Total Station and then export them to CAD or GIS software.

3.4 Methodological Workflow

3.4.1 Planning and Reconnaissance

The survey commenced with a preliminary reconnaissance of the proposed powerhouse area and canal alignment. Our team conducted a field visit to assess site conditions and identify the most feasible route for the canal. Based on ground observations and engineering judgment, the canal alignment was finalized and marked on the field by monumenting the Initial Points (IPs) using wooden pegs. A total of nine IPs were established along the selected alignment to serve as reference points for subsequent survey activities.

3.4.2 Detailing

Following reconnaissance, detailed surveying was carried out using the control points previously established during the traverse survey. A comprehensive topographic survey of the powerhouse site and the proposed canal alignment was conducted. All natural and man-made features within a 25 m corridor on either side of the canal alignment were

recorded to ensure adequate data for design and planning. Additional detailed surveys were performed for other hydropower components, ensuring complete spatial information for engineering analysis.

3.4.3 Data Transfer, Processing, and Digitization

Survey data recorded in the Total Station were exported as a .csv file using the Total Station File Transfer mobile application, enabling efficient data transfer to external devices. The file contained all essential parameters such as easting, northing, elevation, and raw observational data.

The data were imported into Microsoft Excel, where they were automatically arranged into structured columns. Necessary adjustments such as adding appropriate headers and verifying units were made before saving the dataset as a .csv file. This file was subsequently imported into AutoCAD for further processing, where contour generation and topographic mapping were carried out. This workflow ensured accurate, organized, and efficient representation of the survey area for hydropower design purposes.

3.4.4 Contouring, Profile, and Cross-Section Analysis

Using the processed survey data, contour lines were generated to represent the terrain accurately. Longitudinal profiles along the canal alignment and cross-sections at regular intervals were prepared to support hydraulic and structural design. These outputs form the basis for evaluating the feasibility, gradient requirements, and earthwork calculations essential for hydropower development.

4. RESULT

At the initiation phase, the detailed topographical surveying for hydropower around the Jiri Airport area, a closed loop traverse survey was completed to establish a number of control points around that study area which worked as framework for our survey. Altogether of 13 control points were established whose coordinates are shown in table below:

Line	Easting(m)	Northing(m)	Elevation(m)
RO-1000	423958.865	3056888.5	1857.265
1000-3001	423958.865	3056787.54	1852.005
3001-3002	424094.4583	3056200.966	1814.627
3002-3003	424200.7671	3055758.45	1800.056
3003-3004	424147.4655	3055417.999	1795.293
3004-3005	424010.2647	3055436.25	1824.084
3005-3006	424012.4938	3055474.131	1813.251
3006-3007	423963.8114	3055605.122	1810.441
3007-3008	423919.8042	3055901.086	1818.458
3008-3009	423936.8354	3055998.483	1832.828
3009-3010	424017.5248	3056024.021	1815.693
3010-3011	423996.6197	3056348.075	1822.63
3011-3012	423829.1956	3056477.248	1839.226
3012-1000	423797.6312	3056625.285	1852.626
1000-RO	423958.865	3056787.54	1852.005

Table 6: Control Points for Hydropower Surveying

The coordinates of all the Canal Alignment Points are given in the tabulated form as follows:

S.N.	Easting	Northing	Elevation	Remarks
1	423780.268	3056524.862	1853.598	ALT1
2	423814.197	3056414.007	1853.982	ALT2
3	423833.028	3056301.013	1851.77	ALT3
4	423863.477	3056154.106	1850.268	AKT4
5	423793.937	3055952.642	1848.485	ALT5
6	423850.314	3055824.955	1846.689	ALT6
7	423749.425	3056737.796	1857.67	ALT7
8	423941.76	3055380.545	1839.797	ALT8
9	423868.19	3055634.46	1842.111	ALT9

Table 7: Canal alignment points

5. CONCLUSION

Our hydropower survey was done in detail with the control points that had been surveyed using a traverse survey. In this process, longitudinal (L-section) and cross-sections of the proposed canal alignment were drawn. To keep the flow constant; reducing the erosion and sedimentation but at the same time allowing the water to be carried efficiently, a canal gradient of 2 percent was used.

In this survey, we also had a few problems in setting the canal alignment within the necessary gradient. The 2% strict slope ensured that routing was not very flexible especially in steep or irregular terrain areas. Also, some parts of the alignment traversed thick jungle making it physically hard to follow the alignment and impractical in the future construction aspects. These issues highlighted the natural complexity of the implementation of design constraints in natural landscapes.

CHAPTER 3. ROUTE SURVEY AND CURVE SETTING

1. INTRODUCTION

1.1 Background

A route is a strategically laid down path or passageway, which is meant to assist in the flow of people, goods, energy or information. It links two or more places and can be in different forms such as highways, railways, pipelines, power transmission lines and waterways, depending on the purpose of use.

A special form of surveying is route surveying, which is concerned with the exact positioning and location of a route on the ground. It is a methodical gathering, quantification, and examination of information like horizontal and vertical angles, heights, and distances. This data is critical in planning, geometrical design and eventual construction of routes.

Roads and road surveying are the key to successful infrastructure development. An efficient route will improve the connectivity of the region, decrease the time of travel and operation costs, and minimize the negative environmental effects. The route surveying helps in making informed decisions as it determines the most efficient and cost-effective route and thus minimizes the use of materials and construction wastes. It is also safe because it identifies possible risks like unsafe grounds, steep slopes, or natural resources. In addition, route surveying is used to safeguard natural ecosystems, enhance access to remote or underserved areas, and offer important information to urban planning and major infrastructure developments.

Considerations that should be made when choosing route:

- Shortest Possible Length
- Limited Impact on the Environment.
- Economic Feasibility
- Safety and Stability
- Accessibility

1.2 Technical terms used in Route Survey

Longitudinal Profile:

An elevation versus distance curve of the proposed route along the centerline. It is applied to create gradients and cut-and-fill requirements.

Cross-Sectional Graph:

A vertical line that was drawn across the route at intervals. It assists in defining the shape of the terrain, the number of earthworks, and the design of the roadbed and other structures.

Alignment:

The location of the centerline of the route on the ground. The alignment is ideal to provide safety, simplicity of construction and operation, a short path, economy and comfort.

The major influencing factors on the choice of alignment:

- Government standards
- Obligatory points (point to pass and avoid)
- Traffic type and volume
- Geological stability
- Geometric design limits
- Economy
- Terrain stability
- Reducing unwarranted rises and falls.

1.3 Key Curve and Geometry Terms

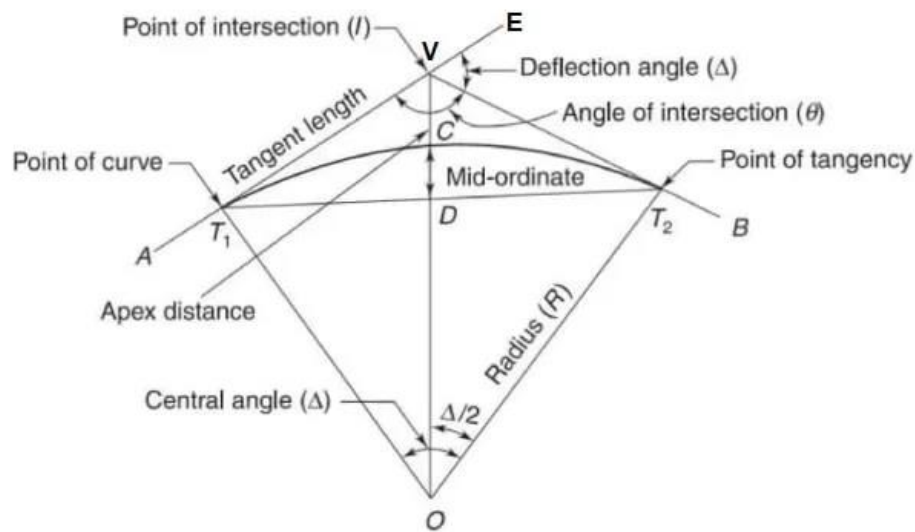


Figure 15: Component of simple curve

Deflection Angle: This is the angle of two tangents at the point of intersection.

Gradient: Increase or decrease with respect to horizontal distance (+ upward, downward).

Chainage Distance traveled along the route using tangents and curves.

Intersection Point (PI): This is the point of intersection of two tangents.

Horizontal Curve: A route which alters the horizontal direction of the route.

Back Tangent: Tangent prior to the commencement of the curve.

Forward Tangent: Tangent at the end of the curve.

Intersection Angle: Angle of two intersecting tangents.

Point of Curve (PC): Start of the curve.

Point of Tangency (PT): Terminal point.

Tangent Distance (T): Distance between PC and PI.

Long Chord: Strauss between PC and PT.

Mid-Ordinate: The distance between the midpoint of the long chord and the curve.

Maximum distance of the curve to the PI (top of the curve).

1.4 Objectives

1.4.1 Primary objective

The primary aim of the study is to gather the specific data about the terrain and physical features along the suggested alignment and create a topographic map of it.

1.4.2 Secondary objective

The study also aims to:

- Identify the most effective and feasible path between the identified points.
- Provide necessary information to aid in planning of geometric design and construction.
- Learn the different uses of the total station (TS) and improve the ability to process and interpret data.

1.5 Scope of work

The focus of this project is the ideas, methods and applications of route surveying, such as the creation of longitudinal profiles and cross-sectional diagrams. It aims to give an insight that can be used in the successful planning and design of transport and utility corridors and take into account technical, environmental and economic factors to facilitate the development of sustainable infrastructures.

1.6 Specification for Route Surveying

Particulars	Specifications
Order of the work	Fourth Order
Purpose	To establish control points and perform a detailed topographical survey.
Instrument	NIKON TS
Unit of Measurement	Linear Measurement: Meter Angular Measurement: Gon

RO to RO tolerance	30 ^{ccg}
Set to Set tolerance	60 ^{ccg}
Face to Face tolerance	2 ^{cg}
Index error tolerance	5 ^{cg}
Linear Misclosure Adjustment	Bowditch Rule
Vertical Misclosure Adjustment	Equal Shift
Bearing Adjustment	Cumulative
Contour Interval	1m

Table 8: Specification fot Link Traverse

1.7 Specification for Curve Setting

Particulars	Specifications
Order of the work	Fourth Order
Purpose	To set out simple horizontal circular curve
Instrument	NIKON TS
Unit of Measurement	Linear Measurement: Meter Angular Measurement: Degree
Instrument Used	Linear Measurement: Measuring Tape Angular Measurement: Total Station
Radius of Curve	30 m
Deflection Angle	60°
Chainage of IP	750 m
Peg Interval	3 m

Table 9: Specification for Curve Setting

2. METHODOLOGY

2.1 Study area

The suitable site that was selected for route survey and simple horizontal curve setting was around Jiri Airport area which is shown in map below:

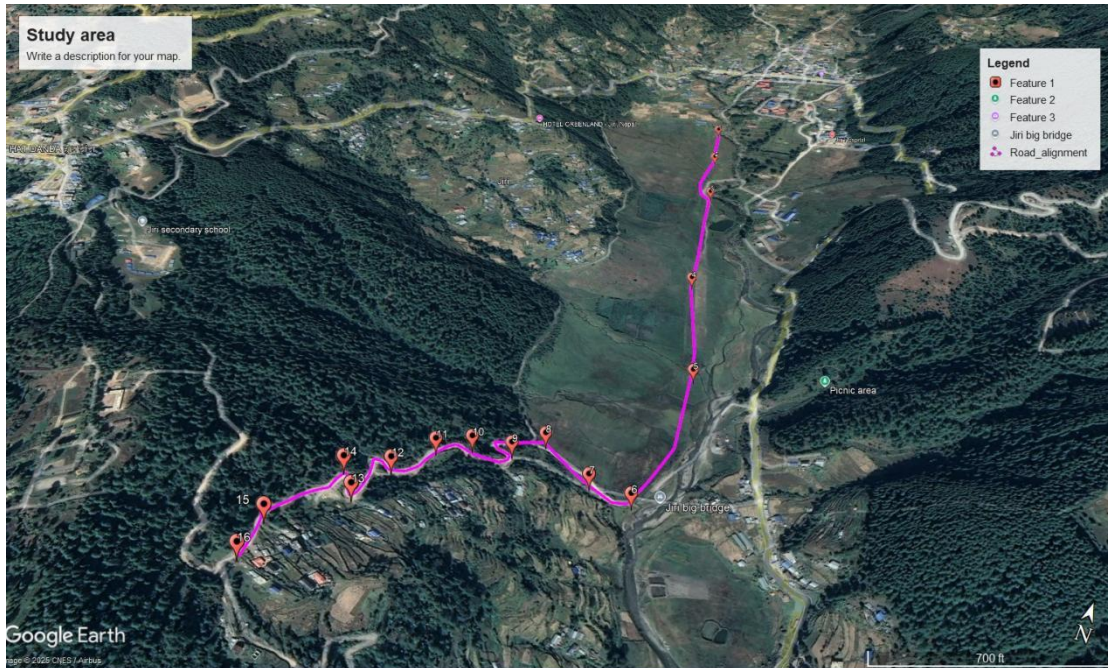


Figure 16: Study area for route surveying

2.2 Instrument Used

- **Total Station**

A Total Station (TS) is a versatile surveying instrument, which is a combination of an electronic theodolite and an electronic distance measurement (EDM) system. It is able to calculate angles, distances and three-dimensional locations and can therefore be used in intricate mapping and construction layouts. The distance measurement is done by use of the EDM that transmits electromagnetic signals to a reflector. These signals are fed back to the instrument and the TS calculates the distance by comparing the phase difference between the outgoing and incoming wave.

- **Tripod**

The tripod is a three-legged stand that is used to firmly hold the total station. The instrument can be adjusted to a comfortable position by its adjustable legs and can be placed on irregular surfaces.

- **Prism (Reflector)**

The electromagnetic signal sent by the EDM of the total station is reflected back by the prism commonly referred to as a reflector. This reflected signal allows the instrument to be able to measure distances accurately.

- **Prism Pole**

A prism pole is a pole that is used to support the prism at a certain height. It has an inbuilt bubble level to make sure that the pole is straight up when making the measurements to enhance accuracy and visibility in the field.

- **Measuring Tape**

Various distances or heights were measured by a 30-meter fiberglass tape, including the height of the instrument (HI).

2.3 Software Used

- MS Excel
- AutoCAD
- ArcMap 10.8.2

2.4 Data Used

The stations were already established from link traverse. Same stations were used for detailing. The coordinates of stations are:

Stations	Easting(m)	Northing(m)	RL(m)
RO1	423926.849	3056804.454	1900.665
1000	423998.434	3056690.833	1890.649
3001	424029.2563	3056422.552	1873.218
3002	424105.5693	3055885.403	1851.3875
3003	424159.1804	3055646.271	1844.8423
3004	424132.1094	3055418.014	1842.073

3005	424070.4596	3055438.158	1853.9466
3006	424000.8365	3055475.904	1859.3608
3007	423968.8243	3055433.213	1873.3015
3008	423944.5753	3055401.361	1882.6439
3009	423894.5675	3055363.695	1897.4049
3010	423877.5381	3055296.309	1906.5459
3011	423843.5705	3055300.446	1913.4183
3012	423867.2009	3055201.972	1929.2627
3013	423832.1456	3055206.044	1937.4228
3014	423821.2711	3055096.742	1958.1514
1001	423822.626	3055052.227	1964.502
RO2	423851.707	3055033.912	1962.837

Table 10: Coordinate used for route surveying

2.5 Methodological Workflow

- Planning

The initial process in a route survey will be to identify the location and establish control points. Possible routes are frequently viewed using Google Earth to check the terrain and determine the ease of access to the area. This early online review is useful in identifying good route options, saves time prior to going to the field and makes planning more efficient.

- Reconnaissance

The detailing work was done on the stations established in the course of the link traverse. This is the initial phase of route surveying which involves a brief inspection of the area to identify potential routes. The primary objective is to gather some general knowledge of the terrain, the shape of the land, and features that might be present.

- Observation and Deflection angles

The points were already in place and thus observations were conducted using the control points. The information such as easting, northing, and the lowered levels of all the existing natural and man-made features were noted down and the spot heights.

Deflection angles are listed below:

Station	Easting	Northing	Elevation	Remarks
3006	424000.8365	3055475.904	1859.2189	CP6
3005	424070.4596	3055438.158	1853.795	CP5
3007	424010.701	3055443.937	1865.476	IP1
3008	423954.345	3055453.081	1865.362	OFF1
3009	423977.64	3055463.218	1862.833	SP1
3010	423990.779	3055457.297	1864.976	IIP1
3011	423998.071	3055446.24	1867.721	IIP2
3012	423997.601	3055436.51	1869.283	IIP3
3013	423982.573	3055427.183	1872.276	IIP4
3014	423975.298	3055412.355	1873.751	IIP5
3015	423998.379	3055375.773	1875.582	IIP6
3016	423997.053	3055367.65	1878.416	IIP7
3017	424032.894	3055324.545	1877.576	IIP8
3018	424044.874	3055296.03	1875.635	IIP9
3019	424039.315	3055267.882	1876.838	IIP10
3020	424045.137	3055233.56	1878.76	IIP11
3022	424057.296	3055227.926	1879.059	IIP12
3023	424069.405	3055214.569	1876.844	IIP13

Table 11: Obtained deflection angles

2.6 Setting Out of Simple Circular Curve

A circular curve can be set out by Linear or chain and tape method or Instrument methods.

We have set out the curves using three different methods:

- Offset from Long Chord
- Offset from Tangent lines
- Offset by Rankine's Method/One-Theodolite Method.
- Offset by Two-theodolite Method

- Offset from Long Chord

This is the linear method of setting out of the simple circular curve. Total station was used for setting out the simple circular curve.

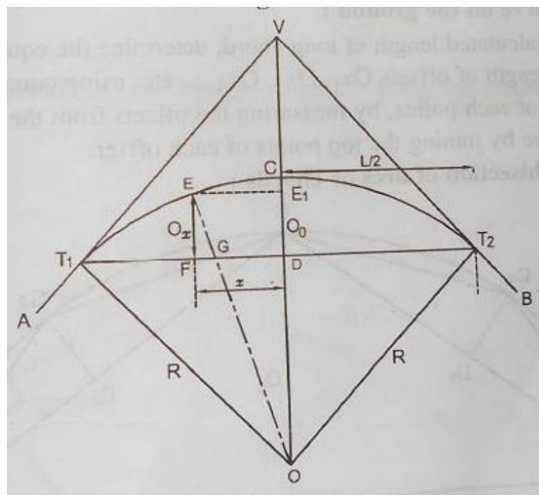


Figure 17: Offset from long chord

As in the figure above T_1CT_2 is the curve to be set between the straight line AT_1I and IT_2B . Let R be the radius of the curve O_0 be the mid-ordinate; O_x be the offset of any point P at a distance x from the mid-ordinate(M). Therefore, the calculations were done as:

From $\triangle OMT_1$: $OM = \sqrt{OT_1^2 - MT_1^2} = \sqrt{R^2 - (L/2)^2}$

$$\text{So, } O_0 = CM = OC - OM = R - \sqrt{R^2 - (L/2)^2}$$

From $\triangle ODD'$: $OD = \sqrt{(OD')^2 - DD'^2} = \sqrt{R^2 - x^2}$

And $OM = OC - CM = R - O_o$

Therefore the offset $O_x = D'P = OD - OM$

$$= \sqrt{R^2 - x^2} - (R - O_0) \dots \text{Exact eqn}$$

$$= R(1-x^2/R^2)^{1/2} - R + O_0$$

$$= R(1 - x^2/2R^2) + x^4/8R^4 - \dots - R + O_0$$

$$= O_0 - x^2/2R \quad (\text{Approx})$$

By assigning various values to x , the corresponding offset values O_x can be calculated. These offsets can then be measured from the long chord, and points can be marked in the field. When these points are connected, they will form the desired curve.

- Perpendicular Offset from Tangent

It is also the linear way of setting out of the simple circular curve. The simple circular curve was set out by use of total station. This technique is appropriate when the radius, length of curve and angle of deflection are small.

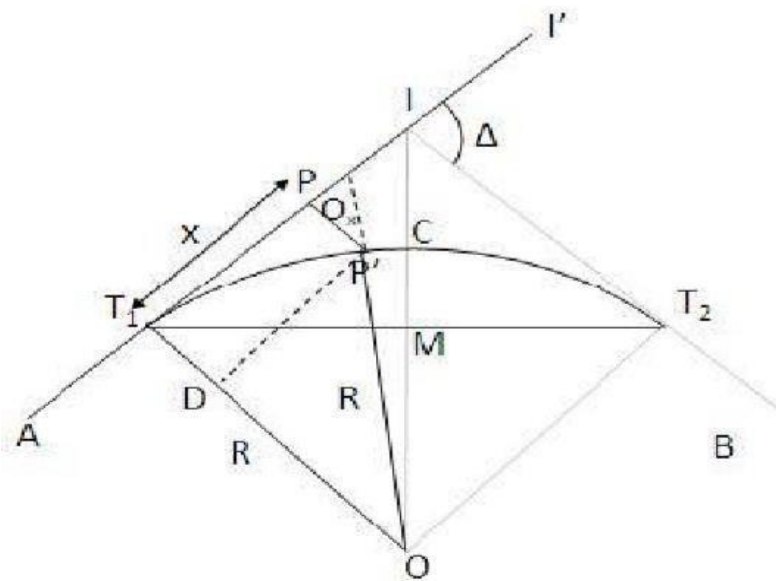


Figure 18: Perpendicular offset from tangent

In the above figure, T_1CT_2 will be the curve to be drawn between the straight lines AT_1 and IT_2B . Where R is the radius of the curve, and O_x is the displacement of any point P at a distance x of the midordinate (M). The calculations were therefore made in the following way:

$$\text{From } \triangle ODP': OP'^2 = OD^2 + DP'^2$$

$$\text{Or } R^2 = (OT_1 - T_1D)^2 + x^2$$

$$\text{Or } R^2 = (R - O_x)^2 + x^2$$

$$\Rightarrow O_x = R - \sqrt{R^2 - x^2}$$

While Approximation,

$$O_x = x^2 / 2R$$

- Rankine Method (One-Theodolite Method)

The technique is useful in drawing a circular curve with long length and large radius. It gives correct values, except where the chords are very long in relation to the radius and there is a considerable difference between the arc length and the chord. It is very dependable and is usually applicable in building of highways and railways.

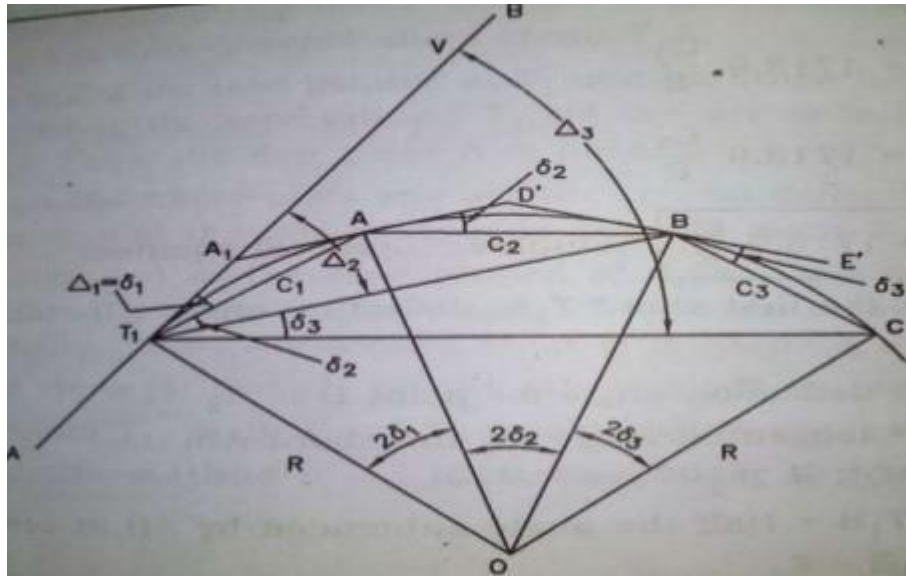


Figure 19: Rankine method

Field procedure:

1. Set out T1 and T2.
2. Set Theodolite at T1.
3. Clamp the plates to the zero point, and point the theodolite at the intersection point.
4. Turn the screw of the upper clamp out and adjust angle Δ_1 0 the Vernier.
5. Holding the end of the tape at T1 and an arrow at a distance $T1A=C1$ swing the tape round T1 until the cross hairs are cut through by the arrow.
6. Slide the top plate aside and adjust the second deflection angle Δ_2 on the Vernier to have the line of sight running along T1B.
7. Fixing the zero end of the tape with a pin at A and, and holding an arrow at a distance $AB=C2$ swing the tape round A till the narrow is cut in half by the cross hairs.
8. Repeat step 6,7 until the final point T2 is achieved.

9. Join the points T1, A, B, C...T2

- Two-theodolite method

This technique is best applied when the ground is not linear and is undulating and rough. Two theodolites are employed in this technique and linear measurements are avoided altogether. Therefore, this is the most precise approach. It is founded on the idea that the angle between the chord and the tangent is the same as the angle that the chord subtends in the adjacent segment (Duggal, 2020). This is the most accurate but costly and time consuming. The different points are determined without depending on the others and therefore the error in determining a point does not pass to the other points.

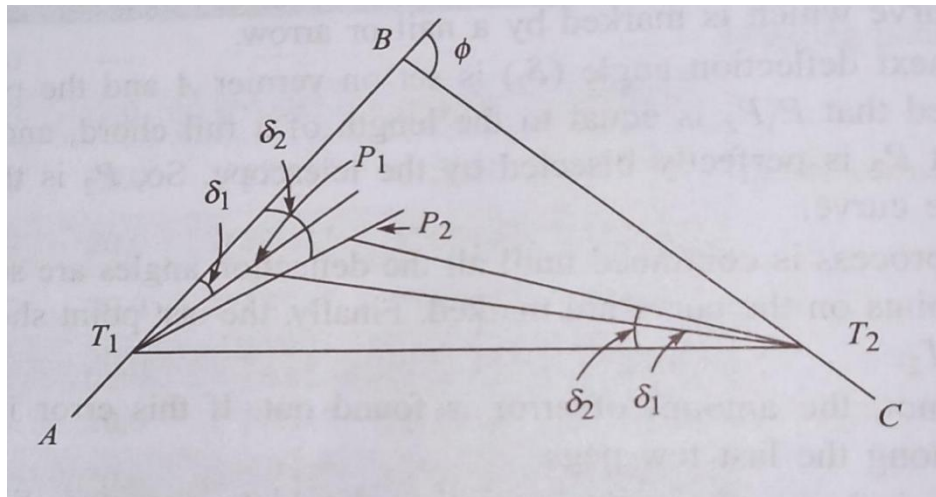


Figure 20: Two theodolite method

Field procedure:

1. Position one theodolite at T1 and the other at T2.
2. Adjust both theodolites to zero Vernier A.
3. Aim the theodolite at T1 at B and theodolite at T2 at T1.
4. Adjust the theodolites to set an angle δ_1 to aim the line of sights at T1P1 and T2P1, thereby establishing the point P1, the intersection points of the two lines of sight, on the curve.
5. Likewise, point P2 is determined by placing δ_2 in both theodolites and by bisecting the ranging rod at P2.

The process is repeated with varying values of δ to define additional points.

- **Data download and plotting data**

The data was downloaded to a phone via the Total Station Data Transfer app, which was subsequently loaded to the computer as an Excel sheet. It was then plotted in AutoCAD.

- **L section and Cross section determination and contouring**

SW_DTM software was used to load the spatial data in AutoCAD. Then it was digitalized. As the data rose, the contours were created at a contour interval of 1m. Then, finally, there was a topographic map with all the cross-sections and profiles.

3. RESULTS AND DISCUSSIONS

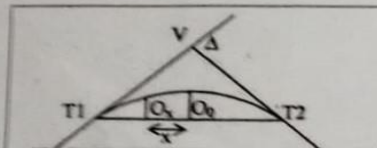
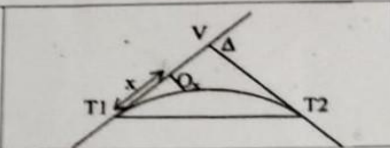
The ultimate outcome of the route survey was to perform the topographic survey of the proposed area with setting out a simple circular curve which was achieved successfully. From the survey we were able to generate a topographic map of the area selected for route with L section and Cross section of the proposed alignment.

Also, the horizontal curve setting out by offset from tangent, Offset from long Chord and Rankine method were done based on the data computed as below:

 **Kathmandu University**
Department of Civil and Geomatics Engineering
Dhulikhel – 7, Kavre

Setting out curves by offset method

Route No: From: To:
 Curve at IP No: Chainage of IP: 1250.500 Date: 20.11.2025
 Deflection angle (Δ): 60° Radius of curve (R): 30m

Tangent Length (T) = $R \tan \Delta/2 = 17.32.00$
 Apex Distance (E) = $R(\sec \Delta/2 - 1) = 4.63.11$
 Chainage of T1 = Chainage of IP - Tangent Length = 1233.18m
 Chainage of T2 = Chainage of T1 + Curve Length = 1264.596m
 Mid-ordinate (O_0) = $R - \sqrt{R^2 - (\frac{L}{2})^2} = 4.63.11$

Offsets from Long Chord

X_5	X_4	X_3	X_2	X_1	X_0	X_1	X_2	X_3	X_4	X_5
15	12	9	6	3	0	3	6	9	12	15
O_5	O_4	O_3	O_2	O_1	O_0	O_1	O_2	O_3	O_4	O_5
$O_x = \sqrt{R^2 - x_1^2} - \sqrt{R^2 - (\frac{L}{2})^2}$					$O_x = \sqrt{R^2 - x_1^2} - \sqrt{R^2 - (\frac{L}{2})^2}$					
0.00	1.514	2.637	3.413	3.868	4.019	3.868	3.413	2.637	1.514	0.00

Offsets from Tangent Lines

X_5	X_4	X_3	X_2	X_1	X_0	X_1	X_2	X_3	X_4	X_5
15	12	9	6	3	0	3	6	9	12	15
O_5	O_4	O_3	O_2	O_1	O_0	O_1	O_2	O_3	O_4	O_5
$O_x = \frac{x^2}{2R}$					$O_x = \frac{x^2}{2R}$					
3.75	2.40	1.35	0.60	0.15	0	0.15	0.60	1.35	2.40	3.75

Observed By: Calculated By:
 Set Out By: Checked By:

Figure 21: Computation form for Offset method


Kathmandu University
 Department of Geomatics Engineering
 Dhulikhel – 04, Kavre

Setting out curves by angular method

Route No: From: To:
 Curve at IP No: Chainage of IP: 1750.500 Date: 2073.11.20.25
 Deflection angle (Δ): 60° Radius of curve (R): 300m

Tangent Length (T) = $R \tan \frac{\Delta}{2}$ = 173.32m Curve Length (l) = $\frac{\pi R \Delta}{180}$ = 31.416m
 Apex Distance (E) = $R(\sec \frac{\Delta}{2} - 1)$ = 4.651m Long Chord (L) = $2R \sin \frac{\Delta}{2}$ = 30m
 Chainage of T1 = Chainage of IP - Tangent Length = 1233.186m
 Chainage of T2 = Chainage of T1 + Curve Length = 1264.596m
 Mid-ordinate (O_0) = $R(1 - \cos \frac{\Delta}{2})$ = 4.82m

Length of first sub chord (C1) = 2.82m Length of last sub chord (Cn) = 0.5954m
 Length of full chord (C) = 4m

Offset by Rankine's Method/ Two Theodolite Method

S.N.	Chord length (m)	Deflection angle (δ_i) = $\frac{1718.87}{R} \times C \text{ minutes}$	Total Deflection Angle (Δ_i)	Angle to be set ($360^\circ - \frac{\Delta}{2} + \Delta_i$)
1.	2.82m	161.57	2° 41' 34"	332° 41' 34"
2.	4m	229.18	6° 30' 45"	336° 30' 45"
3.	4m	229.18	10° 19' 56"	340° 19' 56"
4.	4m	229.18	14° 9' 6"	344° 9' 06"
5.	4m	229.18	17° 58' 17.4"	347° 58' 17.4"
6.	4m	229.18	21° 47' 28"	351° 47' 28"
7.	4m	229.18	25° 36' 39"	355° 36' 39"
8.	4m	229.18	29° 25' 50"	359° 25' 50"
9.	0.5954m	34.11	29° 59' 56.4"	360° "

Observed By: Laxman sharma
 Set Out By: Computed By:
 Checked By:

Figure 22: Computation form for Rankine Method

4. CONCLUSION

The route surveying is an essential procedure in the effective planning, design, and construction of transportation and utility infrastructure. It entails the collection of specific information regarding the terrain, topography, and the features available along the route proposed. The data obtained with the help of the sophisticated equipment, such as the Total Station and other survey methods, assists in identifying the most efficient and cost-effective alignment and reducing the level of environmental and safety issues. The results of the survey, topographic maps, crosssections, and profiles give essential information.

Data on the suitability of terrain to the proposed route, to aid informed decisions on the development of infrastructure. Despite these drawbacks, route surveying is a crucial tool in ensuring the safety, practicability, and sustainability of infrastructure projects, which makes it an invaluable tool in modern engineering and urban development.

CHAPTER 4. BRIDGE SURVEY

1. INTRODUCTION

1.1 Background

Bridges are essential constructions that are constructed to join regions that are divided by rivers, valleys, gorges, or any other natural obstacles. Bridge construction has, over time, developed from the use of simple materials such as wood and stone to the current materials such as steel and reinforced concrete, which are much more durable and effective. Nepal categorizes such structures as culverts (less than 6 m), minor bridges (less than 50 m or less than 25 m), and major bridges (more than 50 m or more than 25 m).

Bridges have been used to facilitate the growth of civilizations by facilitating trade, transportation, and the sharing of knowledge over thousands of years. They are also used as cross-drainage structures, which enhance connectivity in roads and shorten the travel distances.

Bridge surveying is conducted to gather the necessary data on topography, soil conditions, water flow, and traffic requirements to make sure that the planning and design are done correctly. Surveys can be used to identify appropriate bridge alignment through determining the main axis and the other axes, and through evaluation of upstream and downstream conditions. The most important results of the survey are topographic maps, L-sections, and cross-sections of the river.

The bridge survey was conducted in the Jiri River region of Dolakha, where the fourth-order horizontal control points were determined by a Total Station. Critical information like centerline (CL), high flood level (HFL) and flood level (FL), reduced levels (RLs), spot heights and natural and man-made features were documented. These data were subsequently utilized to create detailed sketches, topographic maps and river sections that were needed in the design and construction of the proposed bridge.

1.2 Types of Bridge

There are various ways of classifying bridges depending on their structural form, purpose, alignment, life period, span, deck position and type of superstructure. There are the key classifications as follows.

- **Based on Structural Form**

- **Beam Bridge:** The simplest form of bridge, which is a horizontal beam with abutments on both ends. It supports the weight mainly by bending.
- **Arch Bridge:** Has a curved arch with abutments on both ends. The loads are passed horizontally across the arch to the supports.
- **Suspension Bridge:** The deck is suspended as cables suspended on high towers. These bridges are able to cover a long distance and are anchored by steel pylons or reinforced concrete pylons.
- **Cable-Stayed Bridge:** Cables are attached to one or more towers to directly support the deck. It is appropriate in medium and long spans.
- **Truss Bridge:** Load-carrying Uses truss structure, which is a structure made up of triangular units that are interconnected.
- **Cantilever Bridge:** Made by cantilevers which are projected horizontally and held at one end only.
- **Floating Bridge:** The bridge was supported on pontoons or floats, so that it could rise and fall with the water. Perfect in places where the construction of a bridge is challenging or expensive.

- **Based on Purpose**

- **Road Bridge:** Created to serve car traffic.
- **Railway Bridge:** Constructed purely to serve the trains.
- **Foot Bridge:** Intended for pedestrian use.
- **Aqueduct:** Constructed to pass canal water over barriers.
- **Viaduct:** Transports roads or railways over a valley or rough ground.

- **Based on Life Period**

- **Permanent Bridge:** Built to be used over a long period of time.
- **Temporary Bridge:** Applied in short-term applications like military operations, project implementation or emergency rescue.

- **Based on Span Length**

- **Culvert:** Span less than 6 m.
- **Minor Bridge:** Span between 8 m and 30 m.

- **Major Bridge:** Span between 30 m and 120 m.
- **Long-Span Bridge:** Span greater than 120 m
- **Depending on Location of Bridge Floor**
 - **Deck Bridge:** The topmost level of the superstructure is the roadway/deck.
 - **Semi-Through Bridge:** The deck is positioned in the middle of the superstructure.
 - **Through Bridge:** The bottom is the deck, and the superstructure comes above the deck.
- **Based on the type of Superstructure**
 - Truss Bridge
 - Cantilever Bridge
 - Cable-Stayed Bridge
 - Suspension Bridge
 - Arch Bridge
 - Beam Bridge
 - Tied-Arch Bridge

1.3 Objective

1.3.1 Primary objective

The primary goal of this project was to develop a comprehensive topographic map of the bridge location through repairing the bridge axis and gathering information upstream and downstream required to design the planned bridge.

1.3.2 Secondary objectives

- To locate and repair other bridge axes on both the downstream and upstream sides.
- To choose an appropriate bridge site and alignment depending on the field conditions.
- To undertake in-depth surveys to make longitudinal profiles and cross-sections along the river center-line.

- To gather the required topographical information to aid in the design and construction process.

1.4 Scope of Work

This study aims at undertaking a detailed bridge survey of the Jiri River location in order to create precise topographical data to guide the bridge planning and design. Fourth-order horizontal control points were set up, which implies that the accuracy cannot be higher than fourth-order. The detailed survey was on the presumed river center-line, which spanned 25 meters on each side.

The scope includes:

- Identification of the axis of the bridge and upstream and downstream characteristics.
- The mapping of the topography of the study area in detail.
- Determining the characteristics of river flow, like cross-sectional area and velocity, to estimate discharge (through the flotation method).
- Creating a full set of topographic maps and sections that could be utilized in the future in designing and construction works on bridges.

1.5 Bridge Survey Technical Terms

- **Bridge:** A construction that is made to cross a physical barrier like a river, road or valley to allow vehicles, pedestrians, or utilities to pass through it.
- **Bridge Axis:** This is the supposed center-line of the proposed bridge, which is determined during the surveying stage prior to the commencement of construction.
- **Alternate Axis:** The other potential alignments that would be put forward on both sides of the main axis to determine the most appropriate point to construct the bridge.
- **Upstream:** This is the part of the river that is above the bridge axis.
- **Downstream:** This is the part of the river that is below the bridge axis.
- **HFL (High Flood Level):** The maximum level of the flood that has occurred or is calculated to occur at a given return period.
- **Design Life:** This is the lifespan of a structure that is supposed to be used safely and effectively after construction.

- **Permanent Bridge:** A bridge that is built over a long period of time to ensure that it serves during the whole of the design life.
- **Temporary Bridge:** A bridge constructed temporarily, usually in the course of construction, as an emergency bridge, or as a military bridge.
- **Abutment:** This is the end supports of a bridge, which bear loads of the deck on the ground.
- **Pier:** The vertical supports that are in between the bridge spans.
- **Wing Walls:** Retaining walls that are attached to the abutments and which aid in holding the soil in place and directing the flow of water.
- **Beam and Deck:** This is the supporting part of the load that carries the deck. This is the surface or the floor of the bridge that supports traffic loads.

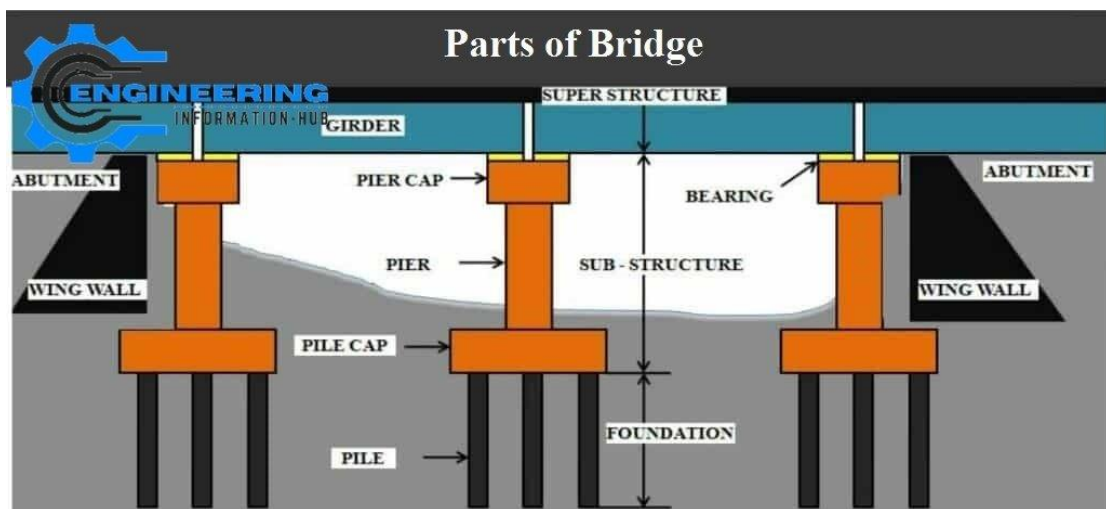


Figure 23: Components of bridge

2. METHODOLOGY

2.1 Study area

The survey area was selected near the Jiri Airpot Area, Jiri Municipality, Dolakha District. The information regarding the discharge of the river was calculated by floating method and High Flood Level were discussed with the local people. The field map of the study area is provided below:



Figure 24: study area for bridge survey

2.2 Material used

- **Total Station**

A Total Station (TS) is a versatile surveying instrument, which is a combination of an electronic theodolite and an electronic distance measurement (EDM) system. It is able to calculate angles, distances and three-dimensional locations and can therefore be used in intricate mapping and construction layouts. The distance measurement is done by use of the EDM that transmits electromagnetic signals to a reflector. These signals are fed back to the instrument and the TS calculates the distance by comparing the phase difference between the outgoing and incoming wave.

- **Tripod**

The tripod is a three-legged stand that is used to firmly hold the total station. The instrument can be adjusted to a comfortable position by its adjustable legs and can be placed on irregular surfaces.

- Prism (Reflector)

The electromagnetic signal sent by the EDM of the total station is reflected back by the prism commonly referred to as a reflector. This reflected signal allows the instrument to be able to measure distances accurately.

- **Prism Pole**

A prism pole is a pole that is used to support the prism at a certain height. It has an inbuilt bubble level to make sure that the pole is straight up when making the measurements to enhance accuracy and visibility in the field.

- **Measuring Tape**

Various distances or heights were measured by a 30-meter fiberglass tape, including the height of the instrument (HI).

2.3 Software Used

- MS Excel
- AutoCAD
- ArcMap 10.8.2

2.4 Methodology

2.4.1 Planning

The bridge location was chosen during the first planning phase following the examination of the High Flood Level (HFL), the stability of the adjacent land and accessibility. The topographic maps of the region were reviewed to learn the topography and hydrology. The location was also selected based on its closeness to the already established control stations in order to have efficient surveying.

2.4.2 Reconnaissance and Monumentation

A reconnaissance survey was made to determine the intervisibility between points, the possibility of traversing the site, and the overall conditions of the site. On the basis of this evaluation, nearest control points from link traverse were chosen, and the consideration of clear line-of-sight, stable ground, and optimum traverse leg set-up was made. Marked

wooden pegs were used to monument offset points and further pegs were put at the proposed bridge axis and other axes that had been identified during reconnaissance.

2.4.3 Fixing the Bridge Axis

The main bridge axis was set at an angle of about 90 degrees to the river bearing in mind the following criteria:

- Both bank stability of soil
- Minimum feasible river span
- Height of bridges in comparison to HF.
- Accessibility to the access road
- A comparison axis was also attached on both sides

2.4.4 Observation

A Total Station was used to conduct traverse observations. The measurements have been made as follows:

- Horizontal distances between traverse stations (two way)
- Both directions Zenith angles
- Horizontal angles of one successive traverse leg to another
- These measurements provided control points with correct positioning and leveling

2.4.5 Detailing

Detailed bridge surveying was done after control surveying along the Jiri River in Dolakha. Information was collected:

- Upstream: 300 m
- Downstream: 200 m

The total station was used to record the following:

- Centerline (CL)
- Left and right river banks

- Reduced levels (RL)
- High Flood Level (HFL)
- Bridge axis & alternative axes
- Spot heights for contouring
- Nature and artificial characteristics.
- The heights were measured at locations with an elevation difference of at least 1 m.

2.4.6 Discharge Measurement (Flotation Method)

River discharge was measured by the flotation method. A floating object (ball) was dropped on the surface of the water and the time that the object took to travel a known distance was recorded.

Steps followed:

1. A place where there was a constant flow of water was chosen.
2. The trapezoidal rule was used to calculate the cross-sectional area, and the depths were taken at 0.5 m depths across the river width (13.358m).
3. The velocity of the flow (v) was calculated as:

$$\begin{aligned} V &= \text{distance}/\text{time} \\ &= 10\text{m}/15.533\text{s} \\ &= 0.6437\text{m/s} \end{aligned}$$

4. The calculation of discharge (Q) was done using:

$$Q = A \times v$$

Where, A = area of cross-section, v = velocity

$$\begin{aligned} &= (13.358 \times 0.1975) \times 0.6437\text{m/s} \\ &= 1.69\text{m}^3/\text{s} \end{aligned}$$

2.4.7 Digitization, Plotting and Data Transfer

The raw data of the Total Station were uploaded in a laptop and converted to CSV format in Microsoft Excel. SW_DTM software was used to import the data into AutoCAD. The

roads, ponds, fences, and other characteristics were digitized with field sketches and recorded coordinates to make sure that there was accuracy.

2.4.8 Preparation of Contours, Profiles, and Cross-sections

Spot heights that were obtained in the field were used to create contours. Longitudinal profiles were drawn along the river center-line, and cross-sections were drawn at a 15-meter interval. These plans played a critical role in the visualization of the terrain variation and bridge design.

3. RESULT

Coordinate values of a main bridge axis point are listed below:

S.N.	easting	Northing	RL
End 1	424044.5	3056420	1825.42
End 2	424094.4	3056453	1825.022

Table 12: Coordinates of the main axis

Length of the main bridge axis is 59.826m.

The coordinates of Alternative axis are:

S.N.	Ends	easting	Northing	RL
1.	End 1	424117.8	3056391	1823.68
	End 2	424067.3	3056366	1823.16
2.	End 1	424126.3	3056264	1814.859
	End 2	424100.2	3056272	1814.855

Table 13: Coordinates of the alternative axis

Length of the alternative axis 1 is 56.251m.

Length of the alternative axis 2 is 27.298m.

4. CONCLUSION

Finally, the bridge survey has allowed preparing detailed topographical maps, river profiles, and alignment options that are necessary to choose the most appropriate bridge axis. The flotation-based discharge estimation was a better insight on the behavior of the river and it facilitated the safe and effective planning of the bridge. These results emphasize the need to take proper field measurements when coming up with durable and functional bridge structures.

We suggest that similar surveys should be conducted in future academic sessions since they enhance practical skills of students and increase their knowledge on surveying methods. This field-based learning is highly beneficial in enhancing competency in the engineering design and construction activities.

CHAPTER 5. STAKE OUT

1. INTRODUCTION

1.1 Background

Staking out coordinates is a key task in surveying where specific ground points are marked according to predetermined map coordinates. In simple terms, staking out means transferring design or map points onto the actual field (Leisch, 1966). It plays a crucial role in ensuring that construction works and other site activities happen in their intended, accurate locations. This process is often referred to as reverse surveying.

During detail surveying, we measure horizontal angle, vertical angle, and distance to obtain the E, N, Z values. In contrast, during stake out, we begin with the E, N, Z values and compute the required horizontal angle, vertical angle, and distance (Bondrea, 2016).

1.2 Principle of Stake Out

The primary purpose of stake out is to accurately shift the design information from the project drawings to the actual ground. This ensures that all construction elements are placed exactly as planned.

1.3 Objectives

The main objectives of stake out is:

- To ensure that each component of the project is correctly positioned in all three dimensions.

The secondary objectives includes:

- To practice and understand the process of staking out coordinates in the field using a total station.
- To apply stake out techniques in engineering and construction layout.

1.4 Scope of Work

The stake out was performed solely for learning purposes. Four random rectangular coordinates were computed and then located on the field using a total station. Therefore, this project is not intended for any practical or future application.

2. METHODOLOGY

2.1 Instruments Used

The following instruments were used during the stake out process:

- i. Total Station
- ii. Prism (Reflector)
- iii. Tripod
- iv. Measuring Tape

2.2 Methodological Procedure

- Staking Out Using Total Station

A suitable station point was chosen to make the stake out process easier, and its coordinates were determined from known control points. When possible, stake out can also be performed directly from the control points.

Using the stake out function in the total station, the device displays the difference between the measured distance and the input stake out distance:

$$\text{Measured Distance} - \text{Stake Out Distance} = \text{Displayed Value}$$

In this operation, one can choose to stake out based on horizontal distance (HD), relative elevation (VD), or slope distance (SD). First, the coordinate of the occupied station is entered. Then the coordinates of the point to be staked out are input. The total station automatically shows the required horizontal angle and distance needed to reach the target point.

The prism is adjusted until the horizontal angle difference becomes zero. Then it is moved forward or backward until the distance difference becomes zero. At that moment, the prism position represents the actual point on the field.

3. RESULTS

All ten assigned coordinates were successfully staked out in the Jiri Airport area with the help of total station. Those ten points are listed below:

S. N	Easting	Northing	Height
1	424099.7	3055872.53	1850.74
2	424107.8	3055854.88	1851.03
3	424108.8	3055855	1850.56
4	424103.6	3055857.81	1850.38
5	424104.6	3055861.89	1851.27
6	424106.4	3055865.24	1851.26
7	424107.4	3055865.66	1850.89
8	424102.3	3055870.05	1851.7
9	424100.8	3055870.1	1851.14
10	424101.8	3055872.48	1851.88

Table 14: Coordinates for stake out

4. CONCLUSION

In field, total of four coordinates were staked out which gave us a clear idea about the reverse surveying. This method also assisted us to get a clear idea about how stake out is done for large scale engineering projects.

CHAPTER 6. PLANE TABLE

1. INTRODUCTION

1.1 Background

Plane table surveying is a graphical surveying technique whereby the field observations and map plotting are performed concurrently. A plane table, which is a drawing board on a tripod, is employed as the measuring instrument and the plotting surface in this technique. A direction sighting is carried out by the use of an alidade and a clinometer can be used to measure elevations. Other accessories like a plumbing fork or U-frame, spirit level, trough compass, drawing sheet and protective cover facilitate correct field operations.

The technique is usually appropriate in small to medium scale mapping, which is usually between 1:10,000 to 1:250,000, particularly in open land where the terrain is readily visible. Plane table surveying is commonly employed to interpolate information between control points determined by more accurate surveying methods, e.g. triangulation or theodolite traversing. Since all measurements are made on the drawing sheet instead of being made in a field book, the surveyor is at the advantage of seeing the landscape and the map that is being drawn at the same time.

The basic rule of plane table surveying is parallelism. According to this principle, the line that is drawn between any two points on the map should be parallel to the line that is drawn between the same points on the ground. The proper orientation of the table, which is normally by positioning it to magnetic north or by resection using known points, is important in ensuring that this parallel relationship is maintained throughout the survey.

Plane table surveying has a number of benefits. As the surveyor plots the observations as soon as he or she sees the terrain, the risks of overlooking significant details are reduced to a minimum, and irregularities can be depicted correctly. Check lines allow errors to be detected and rectified on-site. The technique is fast, less technical in nature and works well in regions where magnetic disturbances cause compass surveying to be inaccurate.

Nevertheless, there are also limitations to the technique. It cannot be used in areas that are densely forested, in wet weather or in areas that need high precision in measurements.

Stability of the setup and the ability of the surveyor to operate the instruments is very important in determining the accuracy of the final map.

In general, plane table surveying is a viable and effective technique of topographic mapping in favorable field conditions with moderate accuracy.

1.2 Objectives

1.2.1 Primary objective

The main aim of the plane table survey was to draw a comprehensive topographical and contour map of the Jiri Airport site, and to directly map the ground information on the drawing sheet. The map was done at 1:500 scale with a contour interval of 0.5 m to reflect the terrain.

1.2.2 Secondary objective

The secondary objectives of the plane table survey were:

- To know and use the old graphical plane table surveying technique.
- To conduct extensive surveying in an efficient and quick way using direct field plotting.
- To determine control points by plane table methods of intersection, resection and by means of auxiliary points.

1.3 Scope of project

This project will entail the drawing of a comprehensive topographical map of the surveyed region by the use of the plane table surveying technique. The mapping was done at a scale of 1:500, and the contour interval was 0.5 meters and which enabled the terrain to be accurately represented.

The fourth-order control points are also established in the survey work, and they are the basis of locating features and providing a reliable plot. Ground details were viewed and drawn on the drawing sheet directly in real time using simple field instruments: the plane table, tripod, drawing sheet, telescopic alidade, plumbing fork, and spirit level.

The scope is also expanded to cadastral surveying, terrain analysis, and data collection to assist in agricultural planning, development of infrastructure, and environmental management. Since plane table surveying allows the visualization of the land

immediately, it is very useful in preliminary surveys, land-use evaluation and large-scale mapping projects where the field mapping is needed in a quick and practical manner.

1.4 Terminologies

1. Plane Table

A drawing board of wood, most commonly of well-seasoned pine or teak, mounted on a tripod. Its top is completely flat and smooth, and the bottom is normally a metallic ring that allows it to be firmly fixed to the tripod.

2. Foresight

Viewed when a station is seen, and a ray is traced to the instrument station that is plotted.

3. Backsight

An observation made by laying the alidade along a line previously traced in the course of the station in hand and another station, and then turning the table until the line of sight falls on the right point on the ground.

4. Resection

One way of finding the position of the plane table station on the drawing sheet by sighting two or more points already drawn on the sheet.

- **Two-Point Problem:** Two known points are used.
- **Three-Point Problem:** Three known points are used to be more accurate.

5. Resection Point (RP)

A special control points the position of which is calculated by establishing the plane table at an unknown point and sighting at least three known stations. These are useful in mapping in detail.

6. Centering

The positioning of the plane table so that the point which is plotted on the sheet is directly over the point which is plotted on the ground, with a plumbing fork or U-frame.

7. Levelling

The procedure of leveling the plane table surface with the help of a spirit level.

8. Orientation

To bring the plane table parallel to the ground lines, that is, to the magnetic north, as it was at the last station.

9. Intersection

One way of finding an inaccessible point is to draw rays to two or more stations whose positions have already been plotted. The point of intersection of these rays is the location of the point.

10. Intersection Point (IP)

An intersection point created by the intersection method, which is typically applied to objects that cannot be accessed to set instruments, e.g. electric poles, spires of temples, roofs of houses, etc.

11. Auxiliary Point

A special control point set (typically at an intersection) to allow the pickup of details in places that cannot be fully seen by the existing stations. In contrast to intersection points, auxiliary points permit instrument set up over them.

12. Plane Table Traversing

A graphical angular measurement (with alidity) and linear measurement (with tape) method of linking a chain of control points to create a traverse network.

13. Great Triangle

The triangle which is made by connecting three known ground points (A, B, C) or their position on the drawing sheet (a, b, c).

14. Great Circle

A circle through the three known ground points (A, B, C) and their respective plotted points (a, b, c) on the map sheet. It is applied in the solution of resection problems.

15. Danger Circle

A case in resection, in which the unknown station is on, or near, the circumference of the great circle. The points in the inner or outer one-third radius of the great circle will result in unstable and inaccurate resection results and should be avoided.

16. Triangle of Error

A tiny triangle is created when rays added at several known points do not converge at one point because of improper orientation of the table, error in observation or plotting errors. The area of this triangle is used to measure the size of the error.

2. METHODOLOGY

2.1 Study area

The study area was selected near the suspension bridge of jiri airport covering some of the residential area.



Figure 25: Study area for the plane table

2.2 Materials and instruments used

The plane table survey involved the following materials and instruments to observe, measure and plot topographic details in the field.

1. Plane Table

A drawing board that is rectangular in shape (usually 75 cm x 60 cm) of well-seasoned pine or teak. The surface on the upper is smooth and flat. The bottom side has a metallic ring that is circular and this is used to connect the board to the tripod with a wing nut. The

board is free to rotate around the vertical axis and can be clamped at any position of choice.

2. Telescopic Alidade

A drawing instrument, consisting of a sighting device, which was employed to draw rays and make inclined sights with more precision than a plain alidade. It consists of:

- Vertical circle telescope (Beamans Arc).
- H-Scale (range: 0–20)
- V-Scale (range: –40 to +40)

Distance Calculations:

- Let I (stadia intercept) = $T - B$
- Horizontal Distance: $D = I \times 100 - I \times H$
- Vertical Distance: $\Delta h = I \times V$
- R.L. of a point: R.L. (station) + H.I $\pm$ $\Delta h - M$

(Where M = middle wire reading)

3. Trough Compass

Adjusted to point the plane table at the magnetic north. It is a freely pivoted magnetic needle with 0 to 5-degree graduations on both ends. The board is in the right position when the needle indicates 0o at both ends.

4. Spirit Level

A bubble is a small metal tube that is used to level the plane table board. The table is also made level by inserting the spirit level in two perpendicular positions and aligning the bubble until it remains in the middle in both directions.

5. Plumbing Fork (U-Fork)

A U-shaped brass frame with:

- A pointer at one end
- A plumb bob is suspended to the other.

- The purpose of this is to center exactly the plotted station on the board over the respective ground point.

6. Plumb Bob

A sharp metallic weight to the plumbing fork. It makes sure that the plane table is accurately centered by simply hanging over the ground station mark.

7. Tripod

A stand is a lightweight but hard stand to support the plane table. It can be folded up and transported. Three leveling screws allow fine leveling.

8. Telescopic Staff

A reading telescopic rod with a length of 35 m that is used to measure the distance and elevation of long distances (more than 100 m). It may be held either vertically or perpendicular to the line of sight.

9. Measuring Tape

Applied in the determination of horizontal distances when picking up details and when setting control points.

10. Compass

To determine direction and help in orienting the plane table to magnetic north where necessary.

11. Perma-Trace Sheets

Durable drawing sheets used because they are:

- Waterproof
- Tear-resistant
- Long-lasting
- Appropriate to be erased again and again.
- They are used as field plotting base materials.

12. Drawing Tools

Other tools that are used in plotting are:

- Pencils
- Inking pens
- Erasers
- Triangular scale
- Paper clips or tape
- Sheet folders

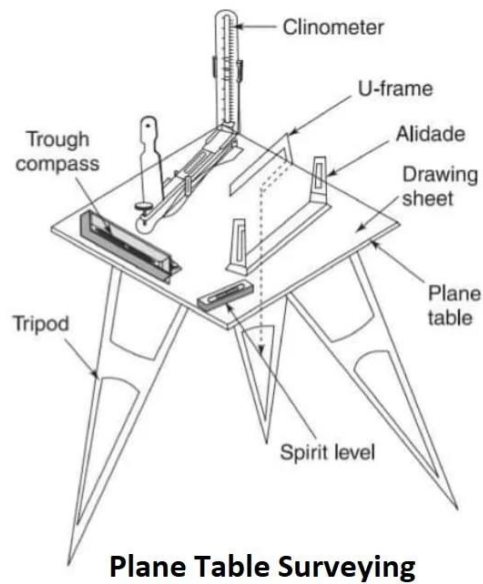


Figure 26: Component of plane table

2.3 Specifications

S.N.	Particular	Specification
1	Order	Fourth order
2	Unit of measurement	Linear: Meters Angular: Degrees
3	Scale	1:500
4	Contour interval	1m

5	Plotting accuracy	12.5cm
6	Purpose	Preparation of topographical map

Table 15: Specification of plane table survey

2.4 Techniques of Plane Table Surveying

There are four primary methods of conducting plane table surveying that can be used based on the visibility, terrain and the intended use of the survey. Such techniques enable the surveyor to find points, define control and map details on the field.

The four major methods are:

- Radiation
- Traversing
- Intersection
- Resection

All of these approaches are described below:

2.4.1 Radiation Method

The most widely and least complicated method of plane table surveying is the radiation method. It is mostly used when the survey area can be seen well at one point.

Procedure:

1. The plane table must be established in an appropriate place where the most ground features can be seen.
2. Plane table Level, centre and orient.
3. Indicate the location of the station in the drawing sheet.
4. Sight every point of detail surrounding the station with the telescopic alidad.
5. Extend rays (lines) of the station point to the detail points.
6. Measure distance, elevation difference (0h) and R.L. with telescopic alidad.
7. Divide the distance measured by the scale of the map and draw the point on the ray drawn.

8. Name the points that have been plotted and connect them accordingly to depict features like buildings, roads and boundaries.

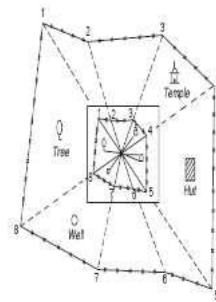


Figure 27: Radiation method

Applications

- Small areas that can be seen by one station are the best.
- Used for large-scale maps (1:500, 1:1000)
- Topographic detail mapping is appropriate.
- Operates in open and flat areas.

2.4.2 Traversing Method

Traversing technique is applied in cases where one station cannot cover a large area. It is like compass or theodolite traversing with the exception that it plots graphically rather than numerically.

Procedure:

1. Choose a sequence of control points which make a traverse.
2. Prepare the plane table at the first station, make it level, centre and face it.
3. See the next station with the alidade and make a foresight line.
4. Determine the distance between the stations by the use of a measuring staff or tape.
5. Take the next station on the foresight on the scaled distance.
6. Go to the next station and repeat the process of orientation by backsighting the last station.
7. Keep on plotting every station until you have traversed it.

8. In case it is a closed traverse, test whether the final point that has been plotted is the same as the first point.

The difference between these points is the closing error, which is to be reduced.

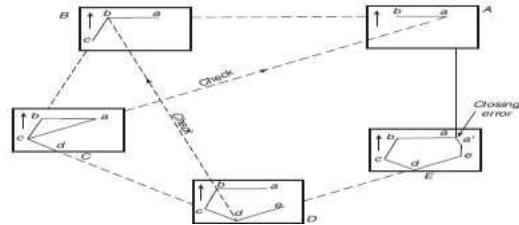


Figure 28: Traversing method

Applications

- Applicable in setting up control points on a long distance.
- Applied to route surveys (roads, canals, boundaries).
- Good in medium-sized areas that need constant connectivity.

2.4.3 Intersection Method

Intersection method applies where the points to be plotted can be seen at two known stations but cannot be accessed or are hard to access. Examples are the roofs of buildings, the tops of hills, electric poles or deep valleys.

Procedure:

1. Take two stations A and B so that all the points desired can be seen.
2. Measure the distance between A and B; this is the baseline.
3. Prepare the plane table at station A and mark point a on the sheet.
4. The table is to be oriented with respect to the ground.
5. View the point P which is inaccessible with the alidade and draw a ray through point a.
6. Go to station B and place the table and face it by backsighting A.
7. Take point P of B once more and draw a ray of point b.
8. The point of intersection of the rays of a and b give the plotted point of P.

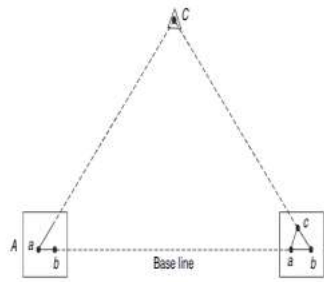


Figure 29: Intersection method

Applications

- Finding objects that are far or inaccessible.
- Surveying rough or hilly areas.
- Used in triangulation of graphs.

2.4.4 Resection Method

When the location of the plane table station on the drawing sheet is unknown, resection is applied to calculate the position of the plane table station on the drawing sheet. The location is calculated by observing familiar, already plotted points.

Three big methods of resection are employed:

A. Back ray orientation

The procedure is performed following the back ray orientation process.

Applied in cases where one of the plotted stations is available and in view.

Procedure:

1. Prepare the table at station A and completely orient it.
2. Take a ray of A to the new station C.
3. Move the table to station C.
4. Take the alidity along the line CA on the sheet, and turn the table until ground bisected A.
5. See the second known point B and make a back ray.
6. The point of the rays determines the location of station C.

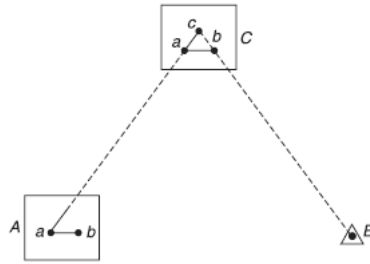


Figure 30: Back ray orientation

Application

- Good where there is only one known point accessible at the new station.

B. Two-Point Problem

Applied when the two known plotted points (A and B) are visible only.

Procedure:

1. Estimate the angle of the table with the line ab.
2. Simultaneously pull back rays to A and B at a guess point.
3. Identify a provisional station D where the rays meet.
4. Go to C and backsight D.
5. Repeat drawback of rays to A and B; rearrange orientation of rays in case they do not intersect.
6. Continue until the C plotted point is correct.

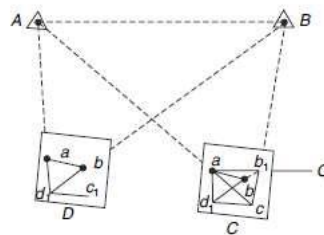


Figure 31: Two point

Limitations

- Less accurate as even minor errors in orientation lead to big positional errors.
- Takes time and needs to be adjusted.

C. Three-point problem

This is the most precise and common technique of resection. A, B, C are known plotted points which are observed at the new station T.

It could be solved in three ways:

a. Mechanical Method of Tracing Paper

- Stick tracing paper on the drawing sheet.
- Indicate a rough value of t .
- Trace rays to A, B and C on the tracing paper.
- Slide the tracing paper until the rays go through the positions of A, B and C that are plotted.
- The sheet is provided with the position t .
- Make sure that the table is well oriented and check back rays.

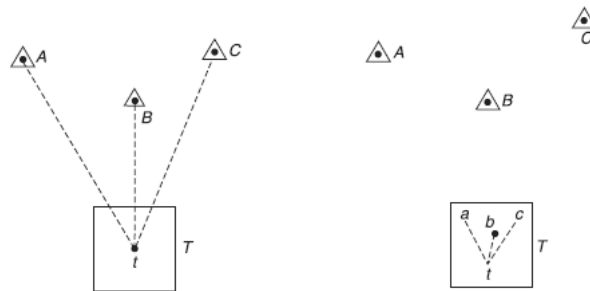


Figure 32: Tracing paper method

b. Graphical Method (Bessel method)

- Set up the table at T.
- Known stations are sighted in a given sequence and intersecting rays are drawn.
- Apply the geometric property of inscribed quadrilaterals.
- The cross over of these rays will provide the precise location of station T.
- The approach by Bessel is favorable since it is fast, graphical, and dependable.

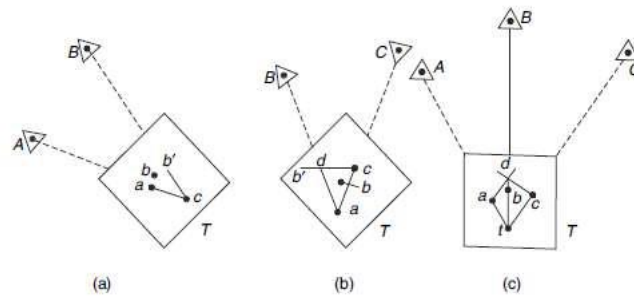


Figure 33: Bessel method

c. Lehmanns (Trial-and-Error) Approach

This is the most popular technique in the field since it enables one to easily rectify errors.

Procedure

1. Determine the position of T on the sheet.
2. Draw rays toward A, B, and C.
3. In case the rays fail to intersect at a point, a triangle of error is present.
4. Determine the position to guess by the rules of Lehmann:
5. The distance between each ray must be proportional to the real distances of A, B and C.
6. The new point should be within the triangle of error.
7. Continue until the triangle becomes a point.

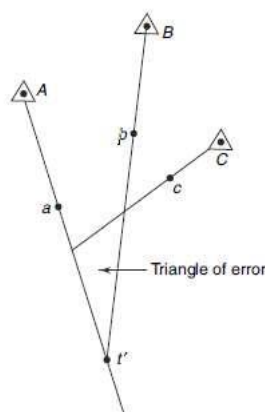


Figure 34: Trial-error method

Applications of Resection

- Applied in cases where position of the station is unknown.

- Assistance in the formation of new control points.
- When changing station in broken country.
- Applied in topographic and cadastral surveys.

2.5 Methodology

- Planning and Reconnaissance

The preliminary steps of the survey are planning and reconnaissance. The survey team together with the instructors (recce) the project area before starting field observations.

Planning Activities

1. Topographical maps and Google Maps of the region that exist were analyzed.
2. Potential and convenient sites of control stations were determined according to the topography and terrain.

Reconnaissance Activities

1. The chosen locations of the stations were checked on the ground.
2. Checking between stations was done.
3. Practical locations of instrument set up and control point locations were verified.

- Monumentation

Monumentation was done after reconnaissance and the finalization of the station location had been done. Each of the chosen control points had wooden or metallic pegs. Signaling was done to make sure that the points were recognizable in the survey.

- Establishment of Control Point

The number of control points was set to 8. It was through traversing methods that a known reference point was fixed. The Total Station observations were used to determine coordinates and elevations of all control stations.

SN	Easting(m)	Northing(m)	RL(m)	Remarks
1	424159.1804	3055646.271	1844.689	Route 3003
2	424225.288	3055561.496	1842.125	Riverside Volleyball

3	424289.732	3055572.008	1868.356	Stairs upward
4	424279.098	3055523.639	1856.138	Roadside above culvert
5	424276.958	3055465.452	1849.526	Middle of bridge and road
6	424318.715	3055382.039	1852.281	Near the suspension bridge
7	424210.141	3055383.213	1839.845	river side below school
8	424132.1094	3055418.014	1841.905	Route 3004
9	424084.182	3055536.91	1845.32	Bridge side beside second electric pole
10	424162.783	3055522.018	1841.789	North side of fence

Table 16: Established point for plane table surveying

- Preparation of a grid sheet / Plotting of the Stations

Plotting was done on a permatrace sheet of 75 cm x 60 cm. The square grid was used to depict an area of 250 m x 250 m at a scale of 1: 500, that is, a 50 cm x 50 cm square grid was used. The square was divided into 10 cm squares vertically and horizontally. The grid and reference coordinates were used to transfer ground coordinates of the control stations to the sheet. To check the stations, a selected point was centered and oriented and rays were drawn to all the visible stations.

- Detailing (Plane Table Surveying)

After plotting the control points, the detail survey was started. Both field observations and plotting were done concurrently. The preparatory steps adopted at each station were as follows:

- Fixing

The plane table tripod was positioned at a comfortable height (approximately at the level of the chest or about 1.2 m).

The butterfly screw was used to clamp the board to the tripod head and clips or drafting tape were used to hold the permatrace sheet in place.

- **Levelling**

A spirit level was used to level off the plane table. The level was laid in two orthogonal directions to make it horizontal.

- **Centering**

The station point on the sheet was plotted at the same point as the ground point. A plumbing fork and plumb bob were used to do this where precision was required. Eye estimation or dropping a small stone through the reference point was used to do rough centering.

- **Orientation**

Orientation was carried out to make sure that the plane table was oriented in the direction that had been established before. The backsight method was employed--the view being taken of the last station and the rotation of the table till the line on the sheet was at right angles to the direction of the ground. This upheld the concept of parallelism between stations.

- **Sighting the Object**

Having fixed, leveled, and centered, and oriented:

1. The station point on the sheet was put on the alidade.
2. The telescope was used to see the object or detail point.
3. A ray was made along the alidade edge, and staff readings were obtained.
4. The horizontal distances, spot heights (SH) and reduced levels (RL) were calculated and indicated on the sheet.
5. The triangular scale was used to plot the natural and permanent features (roads, buildings, trees, boundaries, etc.) by field measurements.

- **Contour Interpolation**

With the spot heights and guide points that have been plotted in the process of detailing: Contours of indices were initially drawn at 5 m intervals. The relief was represented by drawing intermediate contours spaced 1 m apart (or five contours between index contours) using linear interpolation.

- Elements of the Map, Inking, and Layout

Once all the plotting and contouring is complete, critical map features like title, sheet number, scale, north arrow, index, name of cartographer, project coordinator and supervisor information were included.

Special technical pens were used to ink:

- (1) 0.2 mm pen to use in between contours and fine details.
- (2). 0.4 mm index contours, marginal information and title pens.

Before the submission of the map, a final check of completeness and accuracy was carried out.

3. RESULTS

The plane table surveying at the scale of 1:500 and a contour interval of 1 meter was used to prepare topographical map of the work area in Jiri, Dolakha, Nepal. The main aim of the survey was to come up with an elaborate topographical map that will illustrate the key features of the ground including ground elevation, slope characteristics, surface undulations, and natural landmarks. This data can be useful in land-use planning, agriculture and overall land management.

The result from resection is provided below:

Resection point	RL(m)
1	1838.936
2	1843.1512
3	1843.769
4	1842.6142

Table 17: Result of resection points

4. CONCLUSION

Plane table surveying is best suited to small and medium-scale mapping since the observations and plotting are made concurrently, reducing the loss of data. The 5 days of plane table survey made out the topographical map of scale 1:500 with contour interval 1m. The resection and intersection method were used to calculate the location of unknown stations. It enables a rapid analysis on-site and precise depiction of contours and features. The survey must be done with a lot of attention to detail, cooperation, good weather and good instructions by the instructor to have credible results.

REFERENCES

- Awasthi, S. R. (2021). *Renewable energy from small & micro hydro projects: Practical aspects and case studies*. The Energy and Resources Institute (TERI). [AbeBooks](#)
- Boyle, G. (2012). *Renewable energy: Power for a sustainable future* (3rd ed.). Oxford University Press.
- Chow, V. T., Maidment, D. R., & Mays, L. W. (2010). *Applied hydrology*. McGraw-Hill Education.
- Duggal, S. K. (2020). *Surveying* (Vol. 2). McGraw-Hill Education.
- International Renewable Energy Agency. (2021). *Renewable energy statistics 2021*. IRENA.
- Ismail, B. I. (Ed.). (2017). *Renewable hydropower technologies*. BoD – Books on Demand. [Google Books](#)
- Kaunda, C. S., Kimambo, C. Z., & Nielsen, T. K. (2012). Hydropower in the context of sustainable energy supply: A review of technologies and challenges. *ISRN Renewable Energy*, 2012, Article 730631. <https://doi.org/10.5402/2012/730631>
- Korkovelos, A., Mentis, D., Siyal, S. H., Arderne, C., Rogner, H., Bazilian, M., Howells, M., Beck, H., & De Roo, A. (2018). A geospatial assessment of small-scale hydropower potential in Sub-Saharan Africa. *Energies*, 11(11), 3100. <https://doi.org/10.3390/en11113100> [MDPI](#)
- Leisch, A. (1966). *Introduction to surveying*. Prentice-Hall.
- Paish, O. (2002). Small hydro power: Technology and current status. *Renewable and Sustainable Energy Reviews*, 6(6), 537–556. [https://doi.org/10.1016/S1364-0321\(02\)00006-0](https://doi.org/10.1016/S1364-0321(02)00006-0)
- Twidell, J., & Weir, T. (2015). *Renewable energy resources* (3rd ed.). Routledge. <https://doi.org/10.4324/9781315766416>